The LoRaWAN Story: Long Range, Low Power, Tiny Payloads

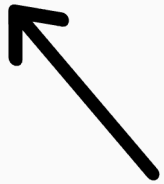
Marco Zennaro, PhD
The Abdus Salam International Centre for Theoretical Physics –Italy

IoT specific case

IoT nodes can accept:

Low throughput, for many applications

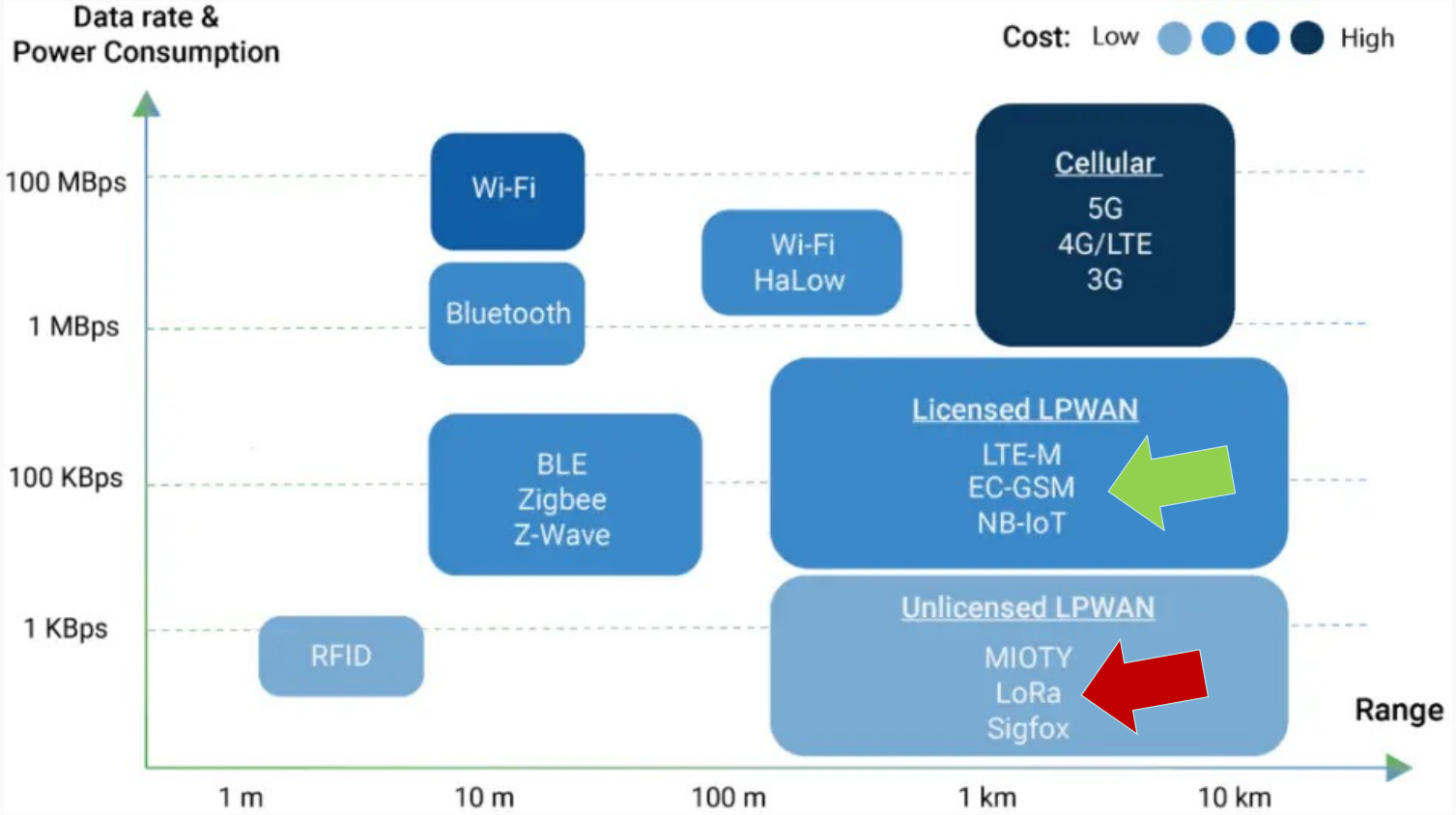
Delays



Which ones don't?

Long Sleeping times

LPWAN: range vs power

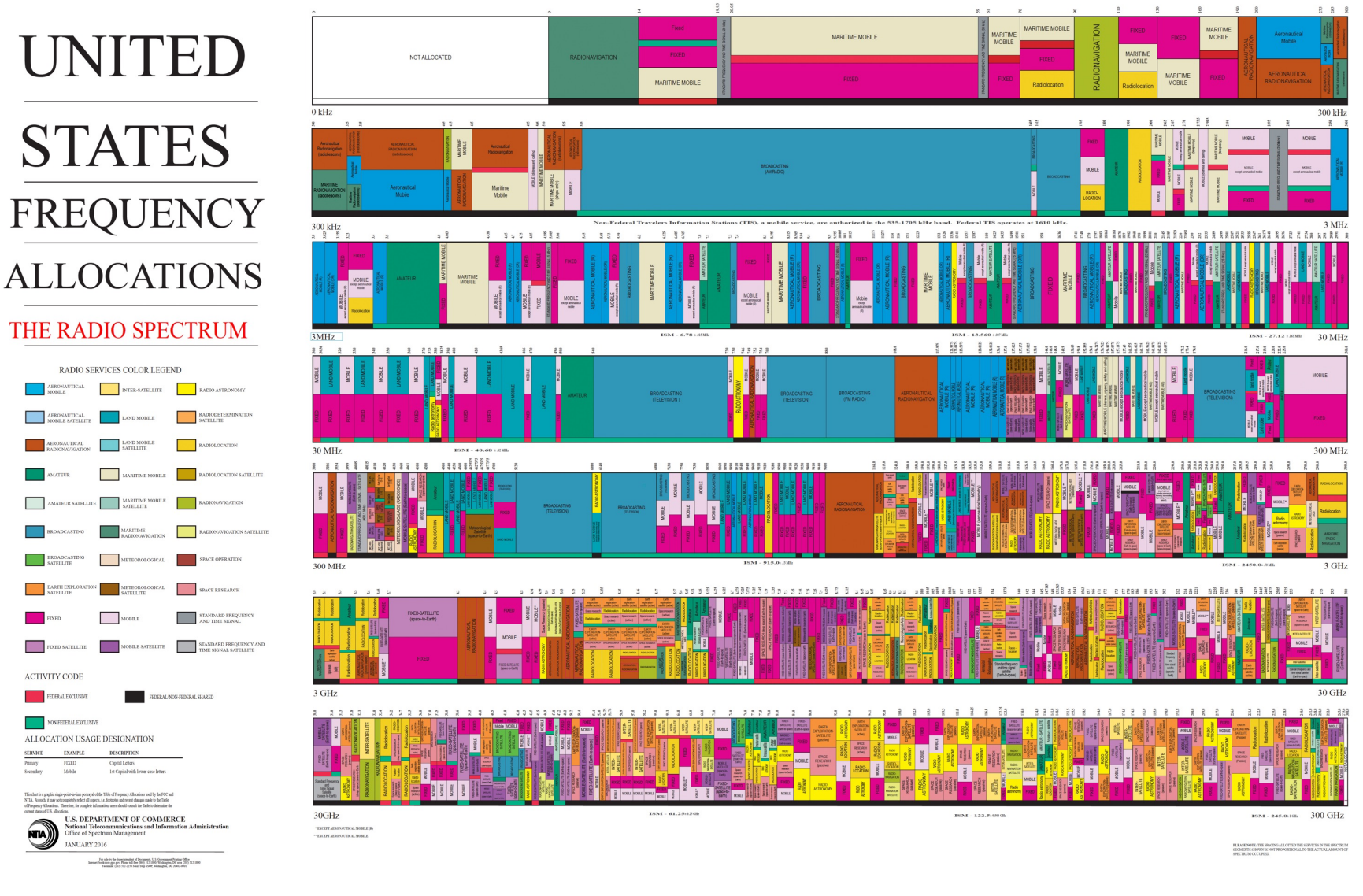


https://www.mokolora.com/lora-and-wireless-technologies/

Something about wireless…



ISM bands


UNITED
STATES
FREQUENCY
ALLOCATIONS
THE RADIO SPECTRUM
RADIO SERVICES COLOR LEGEND
AERONAUTICAL MOBILE
AERONAUTICAL MOBILE SATELLITE
AERONAUTICAL RADIONAVIGATION
AMATEUR
AMATEUR SATELLITE
BROADCASTING
BROADCASTING SATELLITE
EARTH EXPLORATION SATELLITE
FIXED
FIXED SATELLITE
INTER-SATELLITE
LAND MOBILE
LAND MOBILE SATELLITE
MARITIME MOBILE
MARITIME MOBILE SATELLITE
MARITIME RADIONAVIGATION
METEOROLOGICAL
METEOROLOGICAL SATELLITE
MOBILE
MOBILE SATELLITE
RADIO ASTRONOMY
RADIODETERMINATION SATELLITE
RADIOLOCATION
RADIOLOCATION SATELLITE
RADIONAVIGATION
RADIONAVIGATION SATELLITE
SPACE OPERATION
SPACE RESEARCH
STANDARD FREQUENCY AND TIME SIGNAL
STANDARD FREQUENCY AND TIME SIGNAL SATELLITE
ACTIVITY CODE
FEDERAL EXCLUSIVE
FEDERAL/NON-FEDERAL SHARED
NON-FEDERAL EXCLUSIVE
ALLOCATION USAGE DESIGNATION
SERVICE EXAMPLE DESCRIPTION
Primary FIXED Capital Letters
Secondary Mobile 1st Capital with lower case letters
U.S. DEPARTMENT OF COMMERCE
National Telecommunications and Information Administration
Office of Spectrum Management
JANUARY 2016
0 kHz
300 kHz
3 MHz
30 MHz
300 MHz
3 GHz
30 GHz
300 GHz
NOT ALLOCATED
RADIONAVIGATION
MARITIME MOBILE
FIXED
RADIOLOCATION
AERONAUTICAL RADIONAVIGATION
BROADCASTING (AM RADIO)
AMATEUR
BROADCASTING (TELEVISION)
BROADCASTING (FM RADIO)
AERONAUTICAL RADIONAVIGATION
RADIOLOCATION
FIXED-SATELLITE (space-to-Earth)
MOBILE-SATELLITE (space-to-Earth)
RADIO ASTRONOMY
SPACE RESEARCH (passive)
EARTH EXPLORATION-SATELLITE (passive)
ISM - 6.78 ±.015 MHz
ISM - 13.560 ±.007 MHz
ISM - 27.12 ±.163 MHz
ISM - 40.68 ±.02 MHz
ISM - 915.0±.13Mhz
ISM - 2450.0±.5Mhz
ISM - 5.8 ±.075 GHz
ISM - 24.0 ±.125 GHz
Non-Federal Travelers Information Stations (TIS), a mobile service, are authorized in the 535-1705 kHz band. Federal TIS operates at 1610 kHz.
* EXCEPT AERONAUTICAL MOBILE (R)
** EXCEPT AERONAUTICAL MOBILE

ISM bands

Industrial, **S**cientific and **M**edical Radio Bands

ISM Band Frequencies
6.765 - 6.795 MHz
13.553 - 13.567 MHz
26.957 - 27.283 MHz
40.66 - 40.70 MHz
83.996 - 84.004 MHz
167.992 - 168.008 MHz
433.05 - 434.79 MHz
886 - 906 MHz
2.400 - 2.500 MHz
5.725 - 5.875 MHz
24.0 - 24.25 GHz
61.0 - 61.5 GHz
122 - 123 GHz
244 - 246 GHz

f(space, time)

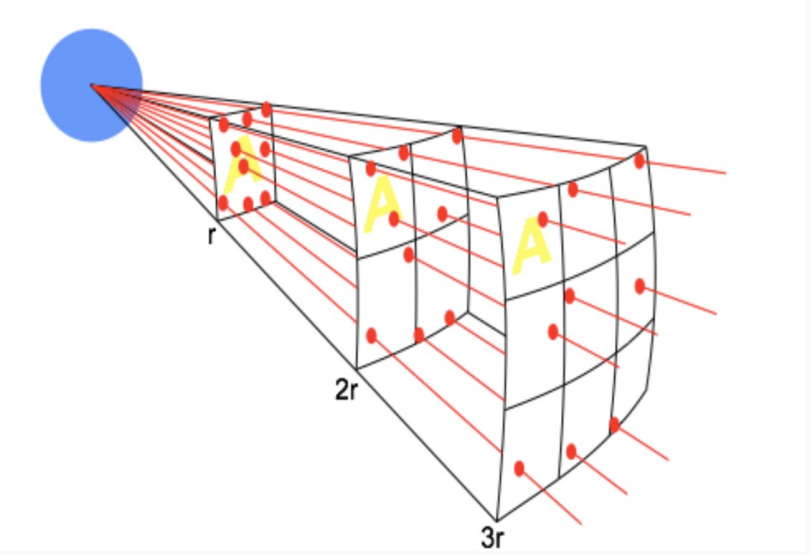
Line of sight (LOS) and free space loss

- Signal power is diminished by geometric spreading of the wavefront, commonly known as ***Free Space Loss***.
- The power of the signal is spread over a wave front, the area of which increases as the distance from the transmitter increases.Therefore, the power density diminishes.
- **Friis free space equation:**

$$\mathrm{FSPL} = \left(\frac{4\pi d}{\lambda}\right)^2 = \left(\frac{4\pi df}{c}\right)^2$$

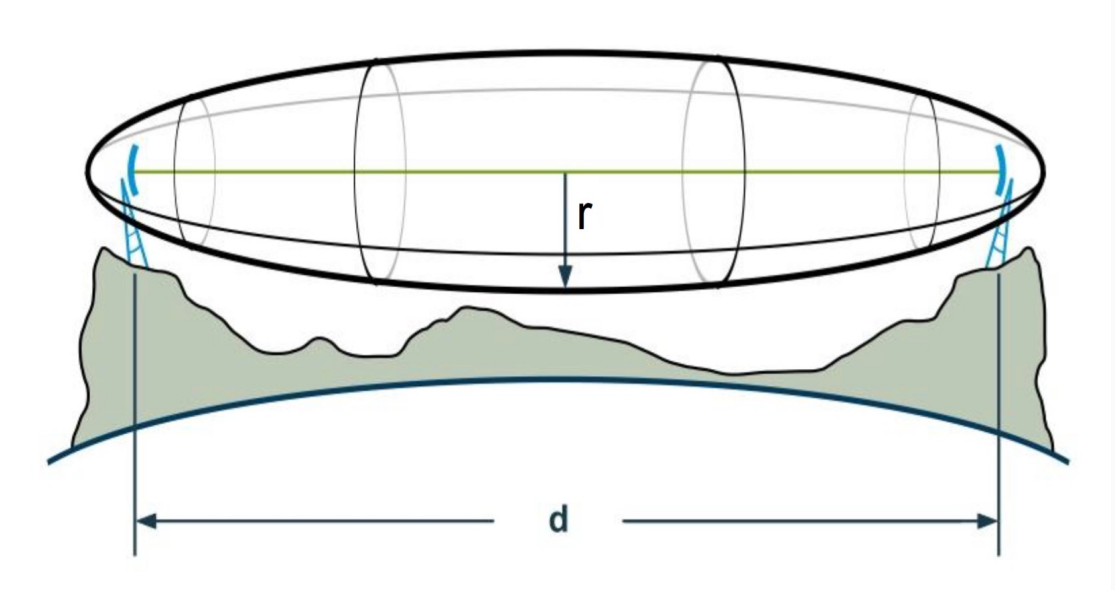
$$L_{fs} = 32{,}45 + 20*\log(D) + 20*\log(f)$$

where L_{fs} is expressed in dB, D is in km and f is in MHz.



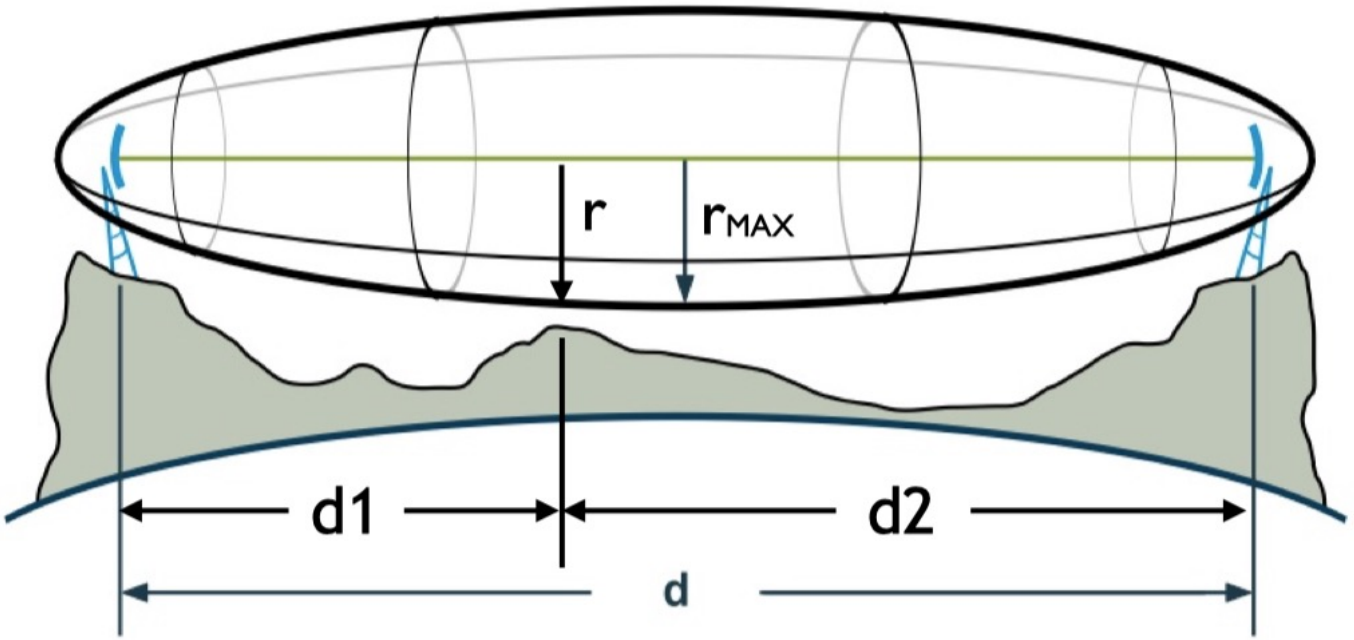
Earth is not flat…and Fresnel Zones

- The First Fresnel Zone is an ellipsoid-shaped volume around the Line-of-Sight path between transmitter and receiver.
- The Fresnel Zone is important to the integrity of the RF link because it defines a volume around the LOS that must be clear of any obstacle for the the maximum power to reach the receiving antenna.
- Objects in the Fresnel Zone as trees, hilltops and buildings can considerably attenuate the received signal, even when there is an unobstructed line between the TX and RX.



http://www.heywhatsthat.com/profiler.html

Line of Sight and Fresnel Zones

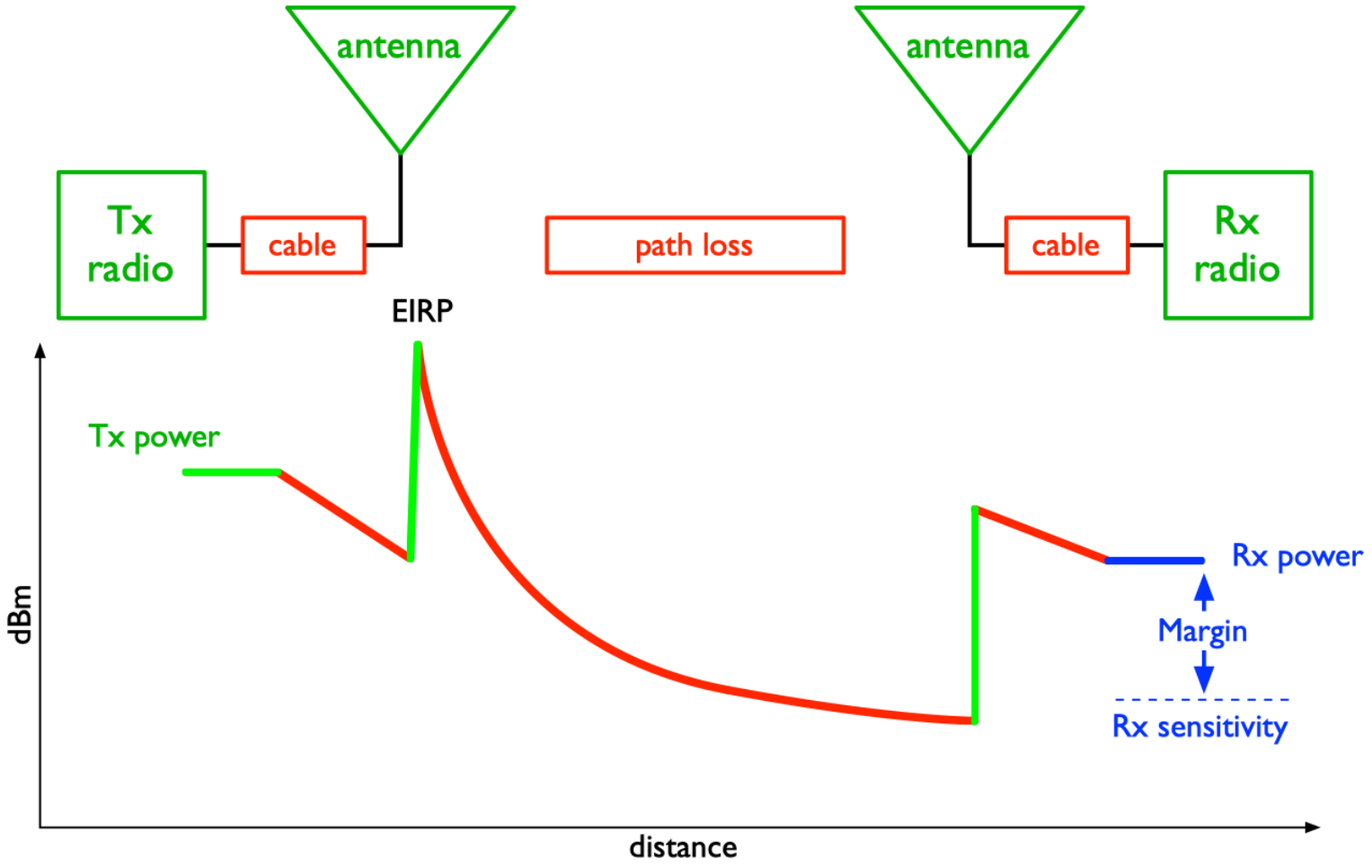


$$r = \sqrt{d1 * d2 * \lambda / d}$$

Link Budget

- The performance of any communication link depends on the quality of the equipment being used.
- **Link budget is a way of quantifying the link performance.**
- The received power is determined by three factors:
  - transmit power,
  - transmitting antenna gain, and
  - receiving antenna gain.
- **If that power, minus the free space loss of the link path, is greater than the minimum received signal level of the receiving radio, then a link is possible.**
- The difference between the minimum received signal level and the actual received power is called the link margin.
- The link margin must be positive and should be maximized

Power in a wireless system



Example

UBIQUITI NETWORKS

BULLET2 DATASHEET

BULLET2

UBIQUITI NETWORKS

Zero Variable Outdoor Wireless Deployment



SYSTEM INFORMATION	
Processor Specs	Atheros MIPS 4KC, 180MHz
Memory Information	16MB SDRAM, 4MB Flash
Networking Interface	1 X 10/100 BASE-TX (Cat. 5, RJ-45) Ethernet Interface

REGULATORY / COMPLIANCE INFORMATION	
Wireless Approvals	FCC Part 15.247, IC RS210, CE
RoHS Compliance	YES

RADIO OPERATING FREQUENCY 2412-2462 MHz

TX SPECIFICATIONS

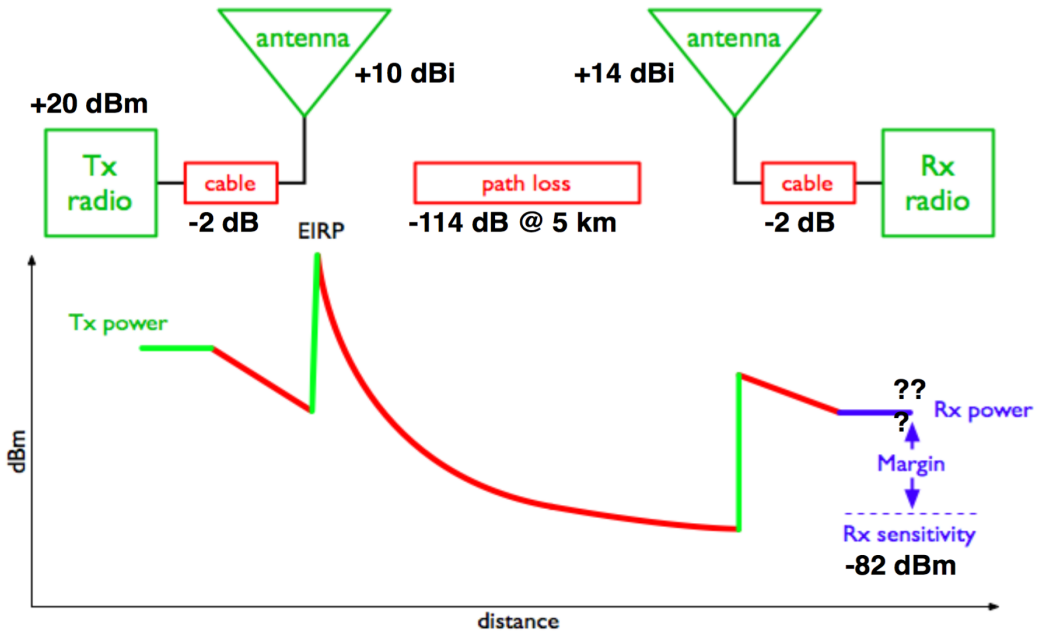
	DataRate	TX Power	Tolerance
802.11b	1Mbps	20 dBm	+/-1dB
	2Mbps	20 dBm	+/-1dB
	5.5Mbps	20 dBm	+/-1dB
	11Mbps	20 dBm	+/-1dB
802.11g OFDM	6Mbps	20 dBm	+/-1dB
	9Mbps	20 dBm	+/-1dB
	12Mbps	20 dBm	+/-1dB
	18Mbps	20 dBm	+/-1dB
	24Mbps	20 dBm	+/-1dB
	36Mbps	18 dBm	+/-1dB
	48Mbps	16 dBm	+/-1dB
	54Mbps	15 dBm	+/-1dB

RX SPECIFICATIONS

	DataRate	Sensitivity	Tolerance
802.11b	1Mbps	-95 dBm	+/-1dB
	2Mbps	-94 dBm	+/-1dB
	5.5Mbps	-93 dBm	+/-1dB
	11Mbps	-90 dBm	+/-1dB
802.11g OFDM	6Mbps	-92 dBm	+/-1dB
	9Mbps	-91 dBm	+/-1dB
	12Mbps	-89 dBm	+/-1dB
	18Mbps	-88 dBm	+/-1dB
	24Mbps	-84 dBm	+/-1dB
	36Mbps	-81 dBm	+/-1dB
	48Mbps	-75 dBm	+/-1dB
	54Mbps	-72 dBm	+/-1dB

Example link budget calculation

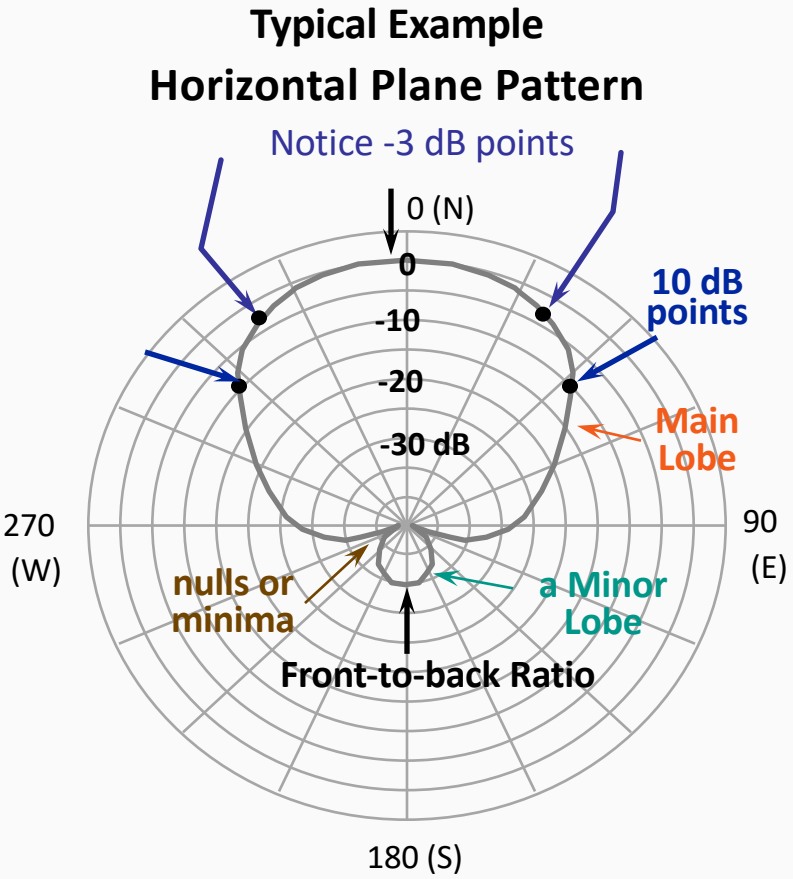
- Let's estimate the feasibility of a 5 km link, with one access point and one client radio.
- The access point is connected to an antenna with 10 dBi gain, with a transmitting power of 20 dBm and a receive sensitivity of -89 dBm.
- The client is connected to an antenna with 14 dBi gain, with a transmitting power of 15 dBm and a receive sensitivity of -82 dBm.
- The cables in both systems are short, with a loss of 2dB at each side at the 2.4 GHz frequency of operation.



20 dBm	(TX Power AP)
+ 10 dBi	(Antenna Gain AP)
- 2 dB	(Cable Losses AP)
+ 14 dBi	(Antenna Gain client)
- 2 dB	(Cable Losses client)
40 dB	Total Gain
- 114 dB	(free space loss @5 km)
- 74 dBm	(expected received signal level)
- (- 82 dBm)	(sensitivity of Client)
8 dB	(link margin)

Radiation Patterns

- An antenna's directivity is expressed as a series of patterns
- The Horizontal Plane Pattern graphs the radiation as a function of azimuth (i.e., direction N-E-S-W)
- The Vertical Plane Pattern graphs the radiation as a function of elevation (i.e., up, down, …)

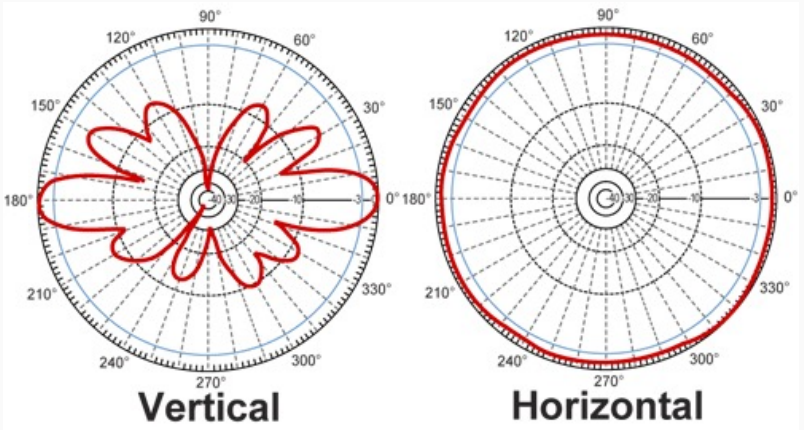


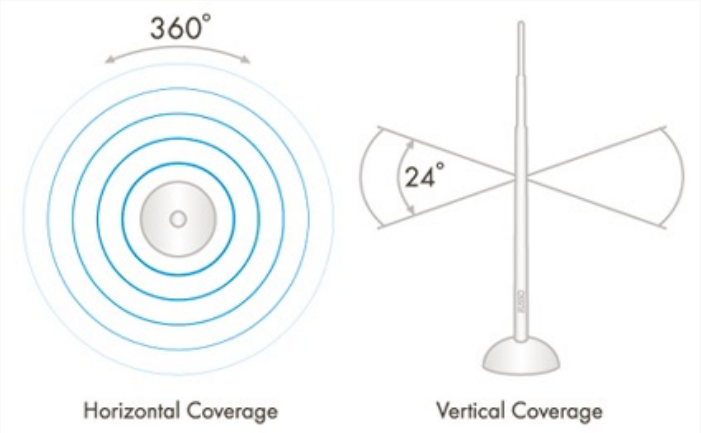
Omnidirectional Antenna

- The smallest, simplest, most practical type of antenna that can be made
  - But that also exhibits the least amount of gain
- Non-directional









Long reach antennas

	PBE-M5-620	PBE-M5-400	PBE-M5-300	PBE-M2-400
Frequency Band	5 GHz	5 GHz	5 GHz	2.4 GHz
Antenna Gain	29 dBi	25 dBi	22 dBi	18 dBi
Dish Reflector	620 mm	400 mm	300 mm	400 mm
Throughput	150+ Mbps	150+ Mbps	150+ Mbps	150+ Mbps
Network Interface	10/100/1000	10/100/1000	10/100	10/100

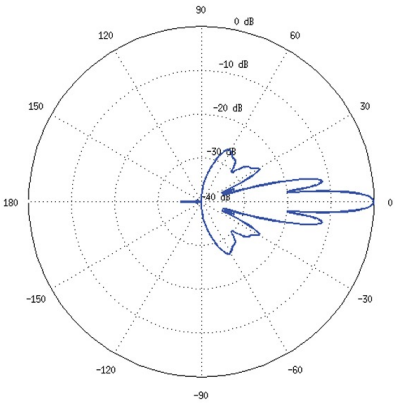


PowerBeam M 400 mm Radome

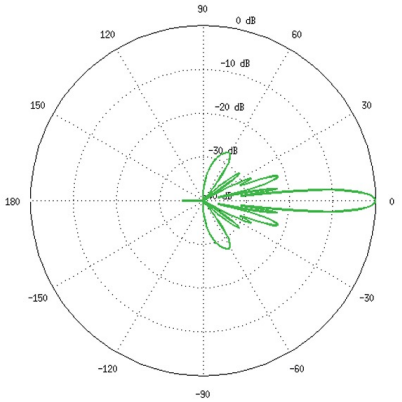
Model	PBE-M2-400	PBE-M5-400	PBE-M5-300
PBE-RAD-400	✓	✓	N/A

A protective radome is available as an optional accessory for the PBE-M2-400 and PBE-M5-400.

E-Plane



H-Plane



Return Loss

LPWAN

What is LoRa

Wireless modulation technology, based on Semtech's proprietary Chirp Spread Spectrum (CSS)

Physical (PHY) layer for long range wireless communications
Operates in the license-free Industrial Scientific Medical (ISM) bands all around the world

Based on spread spectrum, trading bandwidth for S/N.



What is LoRa

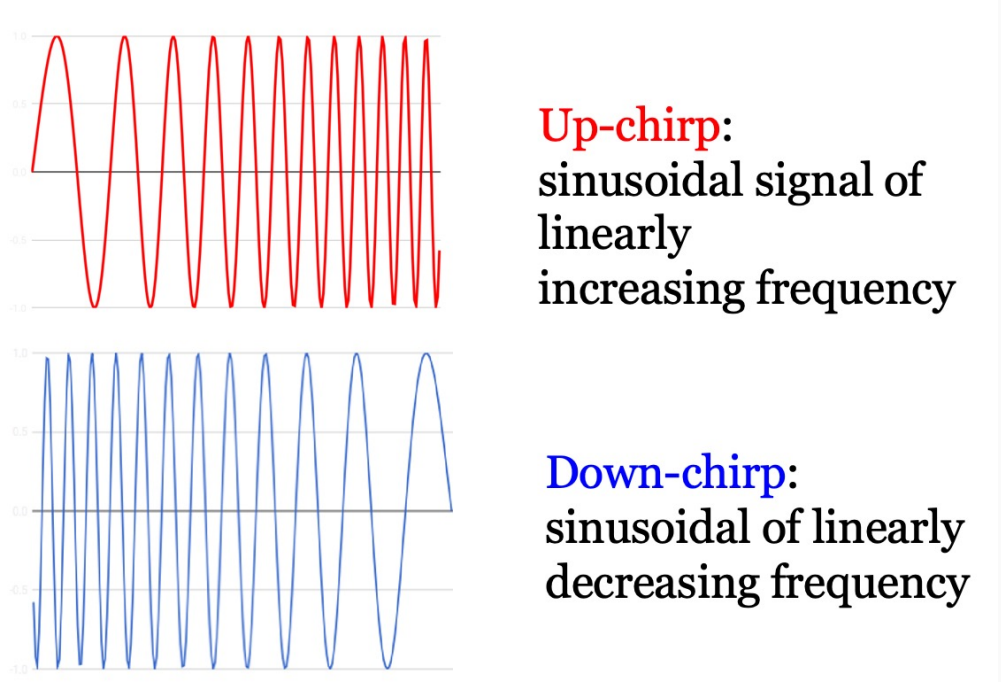
Sub-GHz frequency, e.g: 433, 868, 915 MHz, depends on the country's regulation

Regulated power, duty-cycle, and bandwidth.

E.g: in EU, **1%** per sub-band duty-cycle limitation (per hour, meaning transmission is allowed for 36 sec in each 1 hour)

LoRa modulation

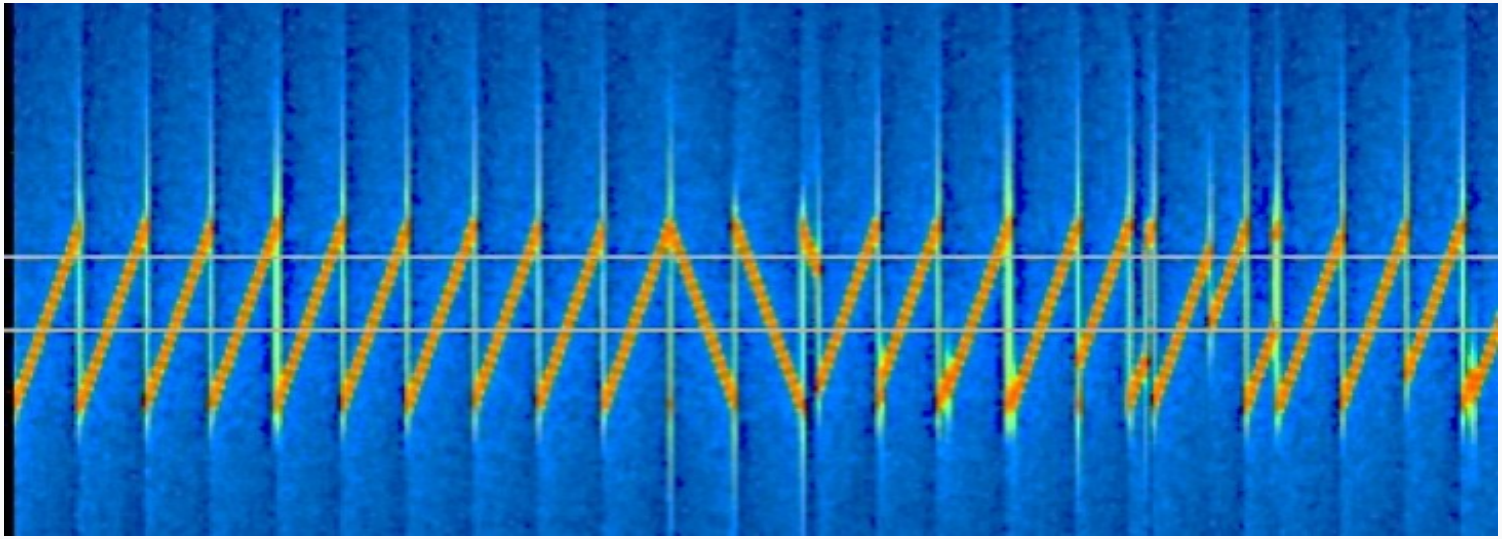
Uses linearly varying frequency pulses called "**chirps**" inspired in radar signals.



Up-chirp:
sinusoidal signal of linearly increasing frequency

Down-chirp:
sinusoidal of linearly decreasing frequency

LoRa Physical Layer

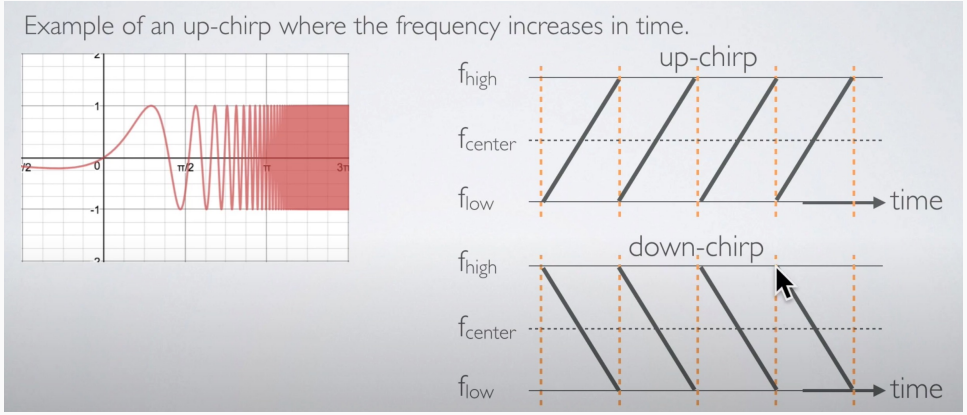


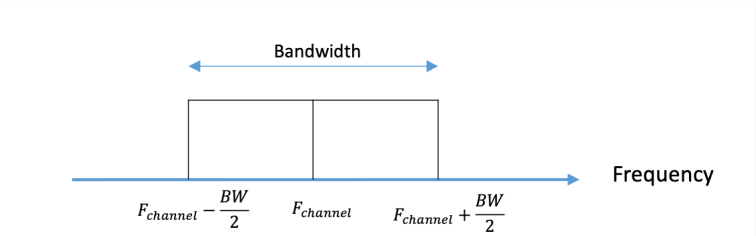
Preamble: at least 10 up-chirps followed by 2.25 down-chirps

Data: information transmitted by the instantaneous frequency transitions

Chirp Modulation

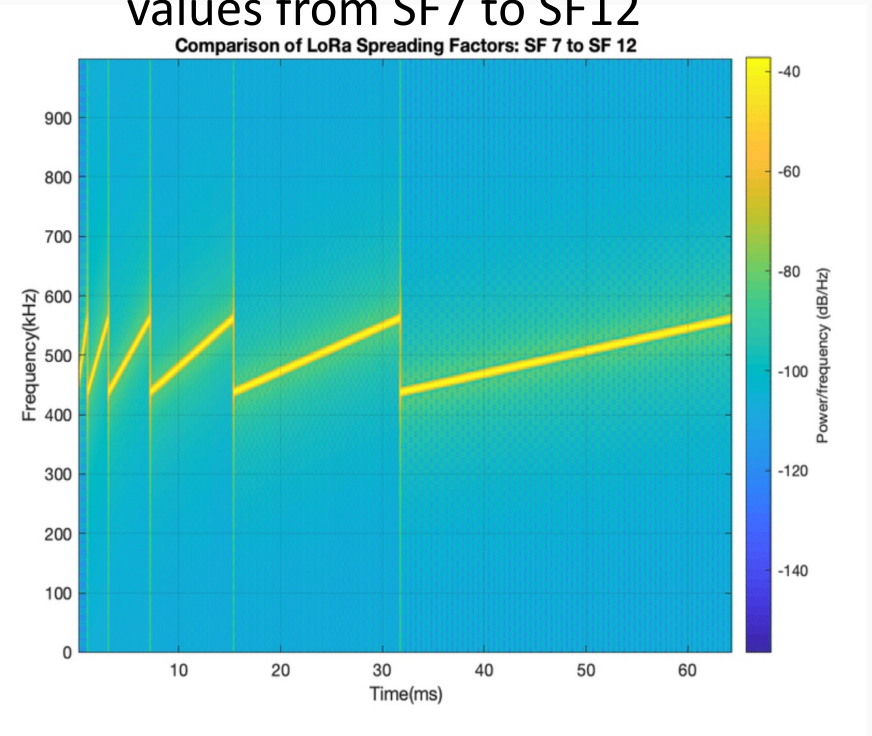
A chirp is a signal in which the frecuencies increases (up-chirp) or decreases (down-chirp) with time





There are 2 critical factors

- **Bandwidth** (BW): 3 possible values 125, 250, and 500 kHz.
- **Spreading factor** (SF): 6 values from SF7 to SF12



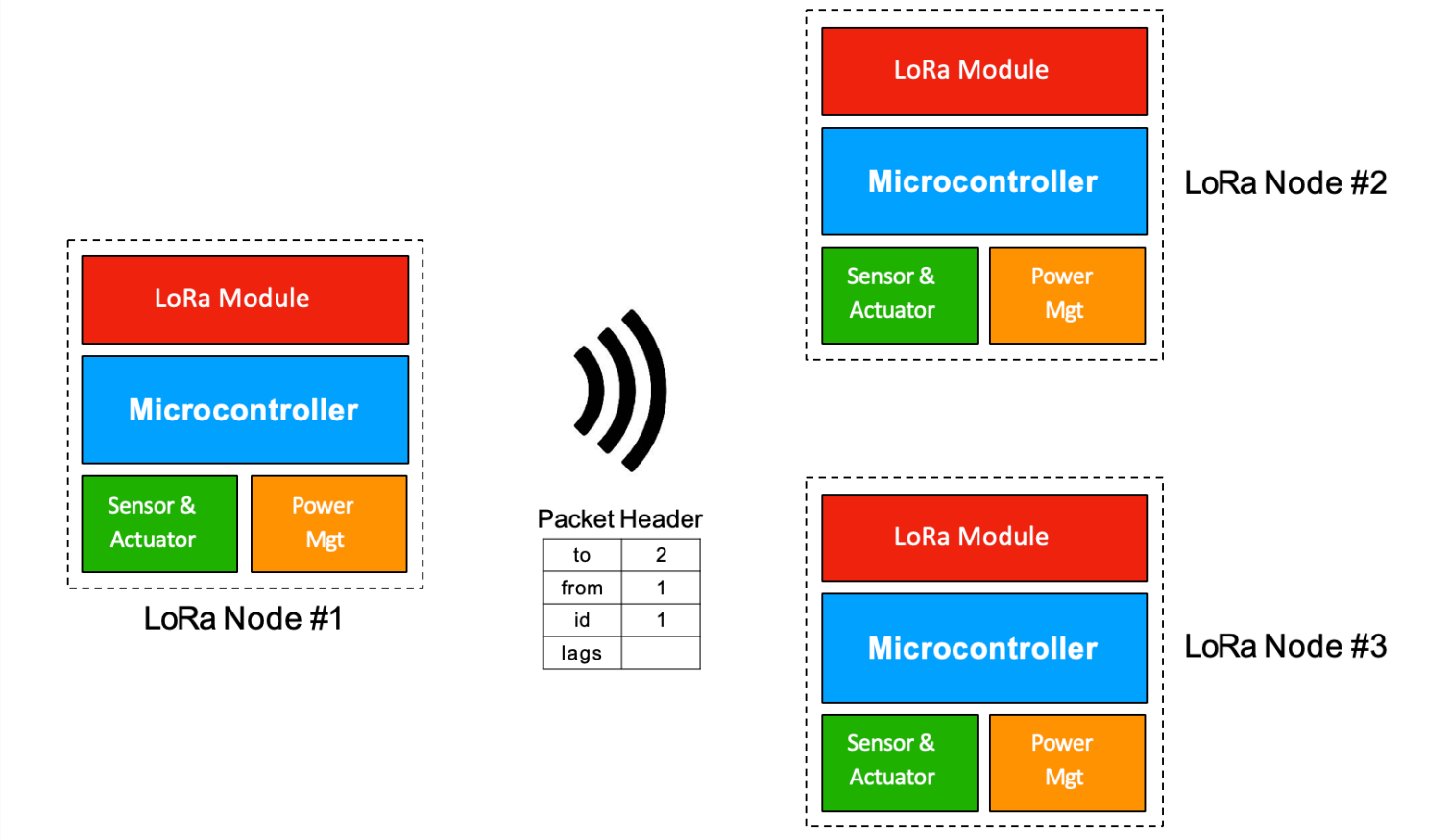
Kim, Dong-Hoon & Lee, Eun-Kyu & Kim, Jibum. (2019). Experiencing LoRa Network Establishment on a Smart Energy Campus Testbed. Sustainability. 11. 1917. 10.3390/su11071917.

What is LoRa

LoRa physical layer consists of many parameters which can be configured into 6720 different settings!

SeRing	Values	DeSnition	ETects
Bandwidth	125, …„500 kHz	Width of spectrum occupied by chirp	A higher bandwidth is required for transmiRing data at high rates (1 kHz = 1kcps). However, increasing this parameter decreases the communication range and sensitivity.
Spreading Factor	2^6, …„2^{12} chips/ symbol	Number of bits encoded per symbol. Symbol is RF state representing some quantity of information. SF12 means 2^{12} chips/symbol, 12 bits of data	A higher spreading factor (*SF*) increases the communication range, radio sensitivity, and the signal-to-noise ratio (SNR). However, energy consumption consequently increases.
Coding Rate	1, …„4 or 4/5, …„4/8	Propoaion of transmiRed bits that caries actual data, as opposed to error correction bits. CR1 · 4/(4+1) = 4/5	Bigger coding rates increase the protection against decoding errors and intecerence bursts at the expense of longer packets, longer air time, and higher power consumption.
Transmission Power	−4, …„20 dBm	Transmission power can be adjusted from −4 to 20 dBm, in 1dB steps. Because of hardware implementation limits, the range is o<en limited to 2 to 20 dBm.	The signal-to-noise ratio is increased by increasing the transmission power at the cost of energy expenditure.
Carrier Frequency	137, …„1020 MHz	CF represents the central transmission frequency used in a band, can be programmed between 137 MHz to 1020 MHz, in steps of 61 Hz.	Lower frequency enables to achieve higher communication ranges for the same transmission power. However, selected CF needs to comply with country's regulation.

LoRa nodes


LoRa Module
Microcontroller
Sensor & Actuator
Power Mgt
LoRa Node #1
Packet Header
to 2
from 1
id 1
lags
LoRa Module
Microcontroller
Sensor & Actuator
Power Mgt
LoRa Node #2
LoRa Module
Microcontroller
Sensor & Actuator
Power Mgt
LoRa Node #3

What is LoRaWAN

Communications protocol and architecture utilizing the LoRa physical layer

Open Source and freely available, specified by LoRa Alliance

Star of Starts Topology.

Nodes connect to multiple gateways

What is LoRaWAN

Adaptive Data Rate (ADR) to improve performance

Built-in multiple levels of security: network or application level encryption, frame counter, etc



LoRaWAN end device/mote

Communicates with LoRaWAN gateways, never directly with other motes.

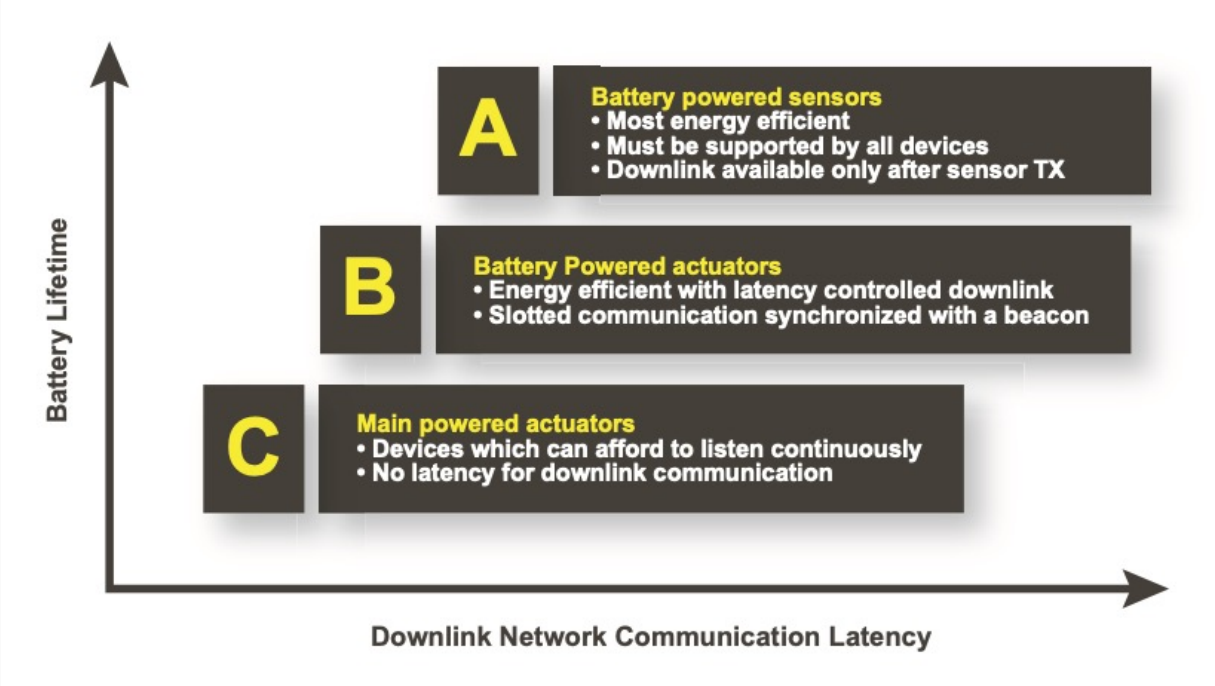
Has 64 bit globally unique identifier: **DevEUI**.

When joining a network, it receives a 32 bit unique identifier: DevAddr.

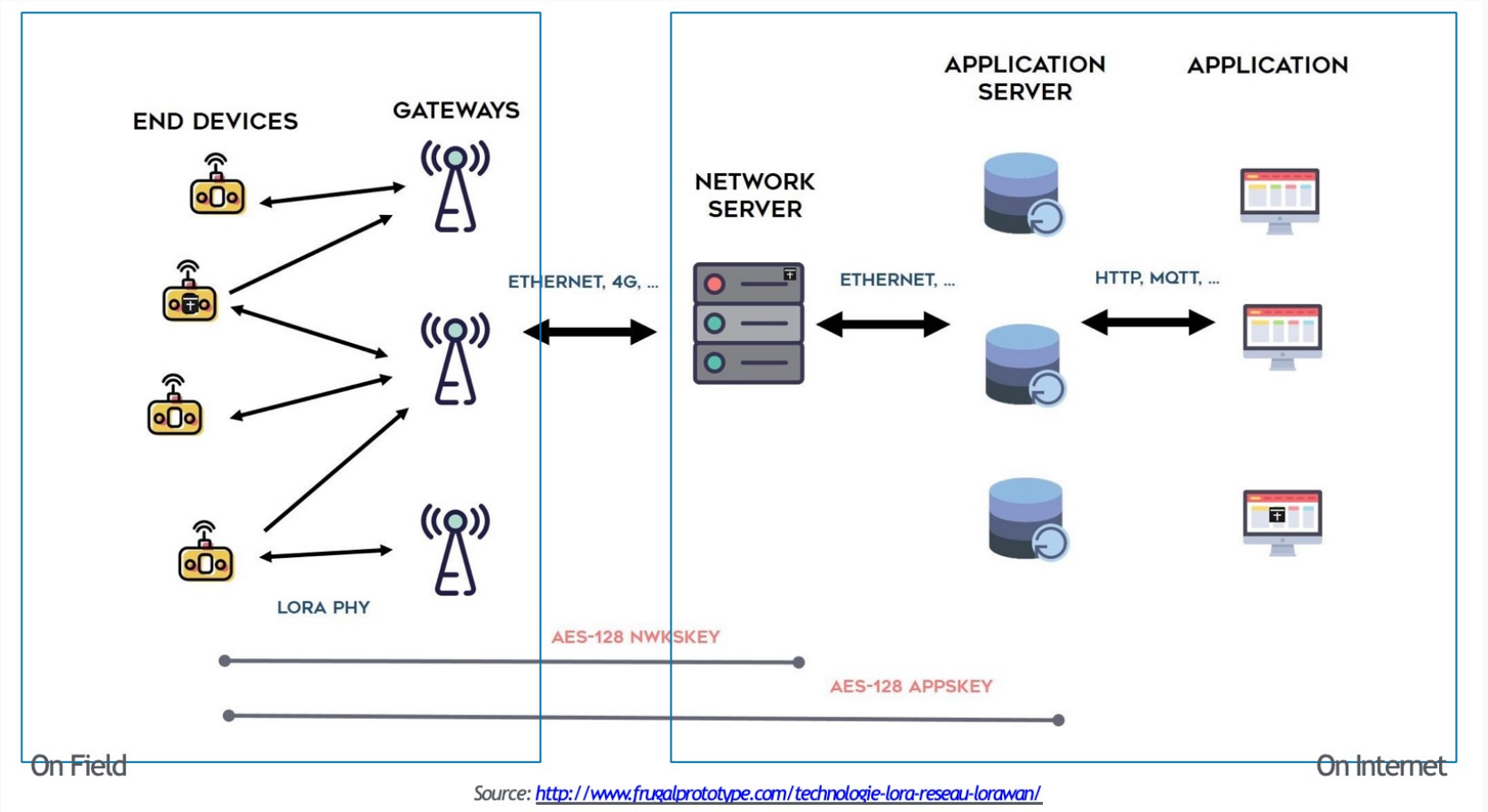
Defined 3 device classes: A, B, and C

LoRaWAN device classes

LoRaWAN has three different classes of end-point devices to address the different needs reflected in the wide range of applications:



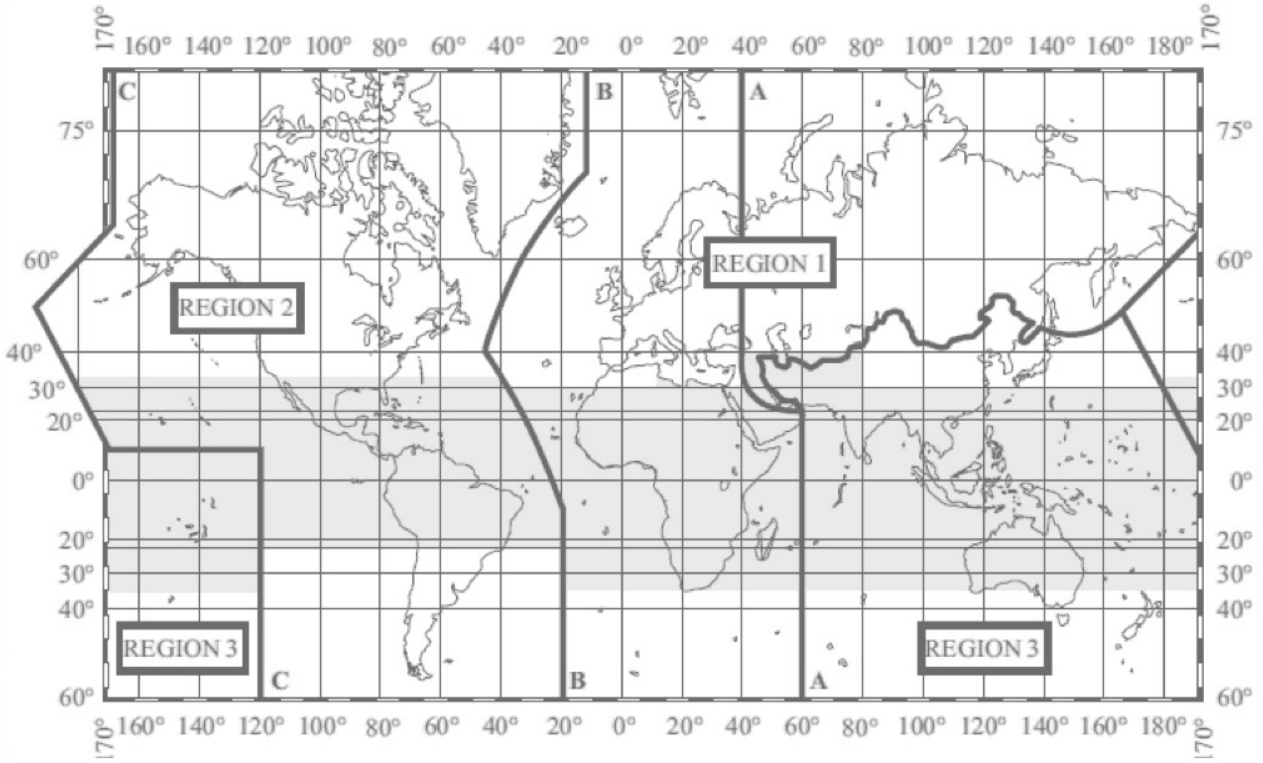
LoRaWAN architecture



Source: http://www.frugalprototype.com/technologie-lora-reseau-lorawan/

LoRa frequency bands

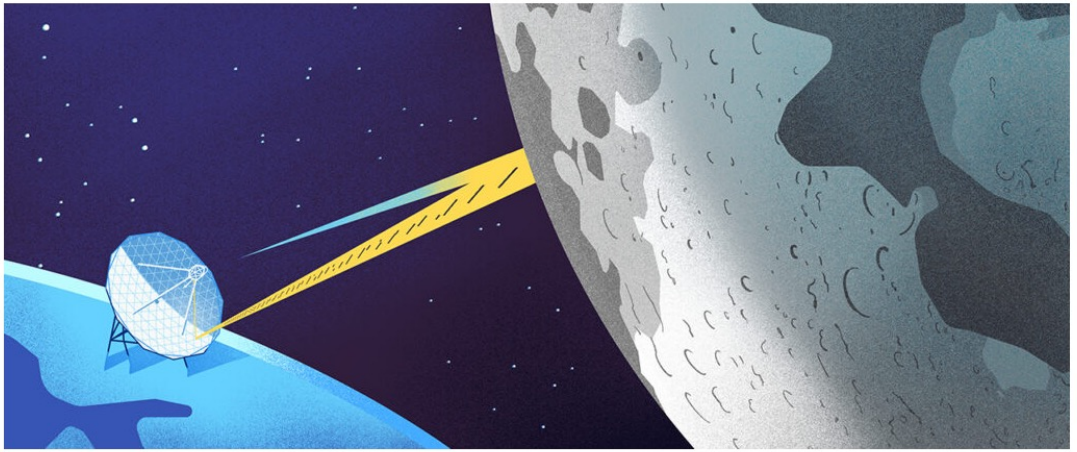
Countries	Frequency band review	Max. output power
EU	868 MHz	14 dBm
USA	915 MHz	20 dBm
Korea	900 MHz	14 dBm
Japan	920 MHz	
Malaysia	862 to 875 MHz	20 dBm
Philippines	868 MHz	
Vietnam	920 to 925 MHz	
India	865 to 867 MHz	
Singapore	922 MHz	
Thailand	920 to 925 MHz	
Indonesia		
ANZ	915 to 928 MHz	
Taiwan	920 to 925 MHz	
China	470 to 510 MHz	17 dBm



LoRa capabilities

First LoRa® message bounced off the moon

24 November 2021



For the first time ever we bounced a LoRa® message off the moon on October 5th 2021, using the Dwingeloo radio telescope. This first was achieved by a team consisting of Jan van Muijlwijk (CAMRAS, PA3FXB), Tammo Jan Dijkema (CAMRAS), Frank Zeppenfeldt (ESA, PD0AP) and Thomas Telkamp (Lacuna Space, PA8Z). The signal traveled an amazing distance of 730,360 km, which to our knowledge is the furthest distance a LoRa® modulated message has ever traveled.

For a short moment the entire message was in space, in between the Earth and the Moon. We transmitted the signal with a Semtech LR1110 RF transceiver chip (in the 430-440 Mhz amateur band), amplified to 350 Watt, using the 25 meter dish of the telescope. Then, 2.44 seconds later, it was received by the same chip. One of the messages even contained a full LoRaWAN® frame.

Influence of Spreading Factors

- **Data rate**
  - A lower spreading factor provides a higher bit rate for a fixed bandwidth and coding rate.

Case 1: For SF7, 125 kHz	>	bit rate = **6,836 bps**
Case 2: For SF12, 125 kHz	>	bit rate = **366 bps**

$$Bit\ Rate = SF.\frac{Bandwidth}{2^{SF}}$$

- **Time-On-Air**
  - Compared to a lower spreading factor, sending a fixed amount of data (payload) with a higher Spreading Factor and a fixed bandwidth needs longer time-on-air.
- **Receiver Sensitivity**
  - Higher spreading factors provide higher receiver sensitivity.
- **Battery Life**
  - The battery life of an end device is highly dependent on the spreading factor used. Higher spreading factors result in longer active times for the radio transceivers and shorter battery life.

Coding Rate

- The Coding Rate is a ratio that increases the number of bits transmitted in order to carry out error detection and correction. In the case of a CR = 4 / 8, there is 8 bits truly transmitted each time we want to transmit 4 bits.

CodingRate (RegModemConfig1)	Cyclic Coding Rate	Overhead Ratio
1	4/5	1.25
2	4/6	1.5
3	4/7	1.75
4	4/8	2

- For example
  - For SF7, 125 kHz and CR4/5 → bit rate = 6.836 kbps / 1.25 = 5469 bps
  - For SF12, 125 kHz and CR4/5 → bit rate = 366 bps/1.25=293bps

- **The documentation of a LoRa transceiver gives the data rates according to the Spreading Factor, the Bandwidth and the Coding Rate.**

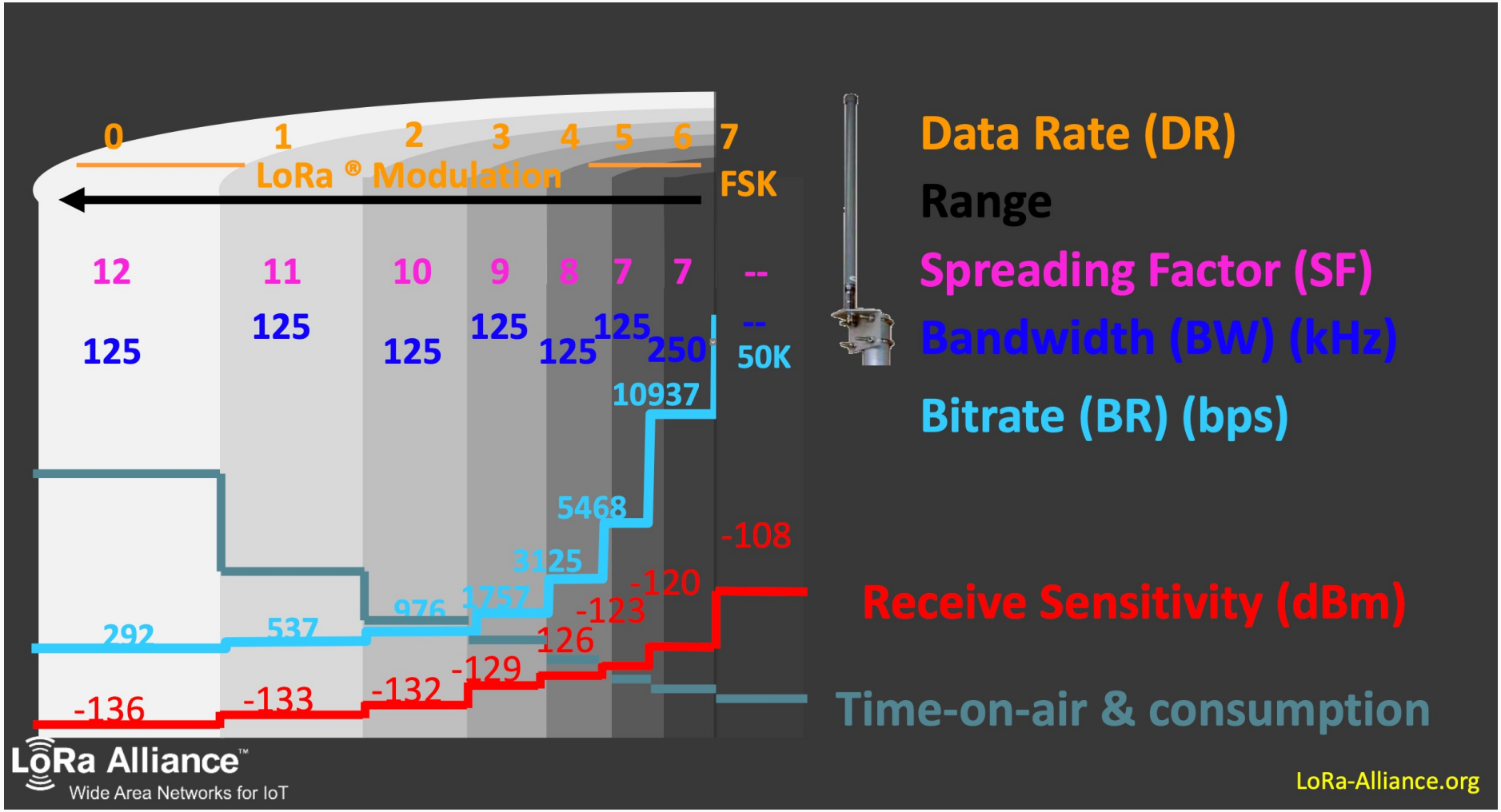
- An online LoRa simulator is also available at https://loratools.nl/#/airtime

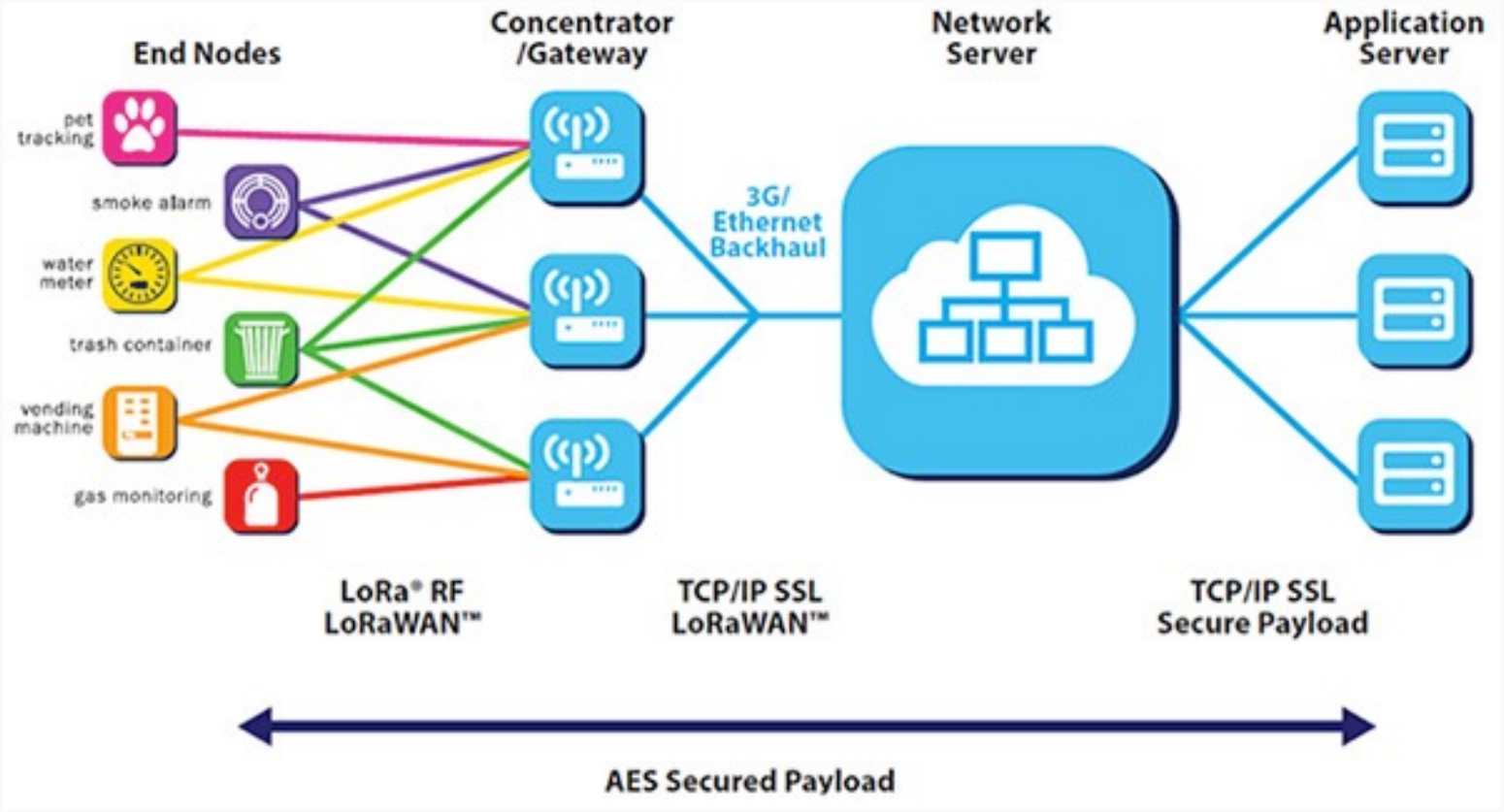
EU863-870 Data Rates

Data Rate	Configuration (SF + BW)	Bit rate (bit/s)
0	LoRa: SF12 / 125 kHz	250
1	LoRa: SF11 / 125 kHz	440
2	LoRa: SF10 / 125 kHz	980
3	LoRa: SF9 / 125 kHz	1760
4	LoRa: SF8 / 125 kHz	3125
5	LoRa: SF7 / 125 kHz	5470
6	LoRa: SF7 / 250 kHz	11000

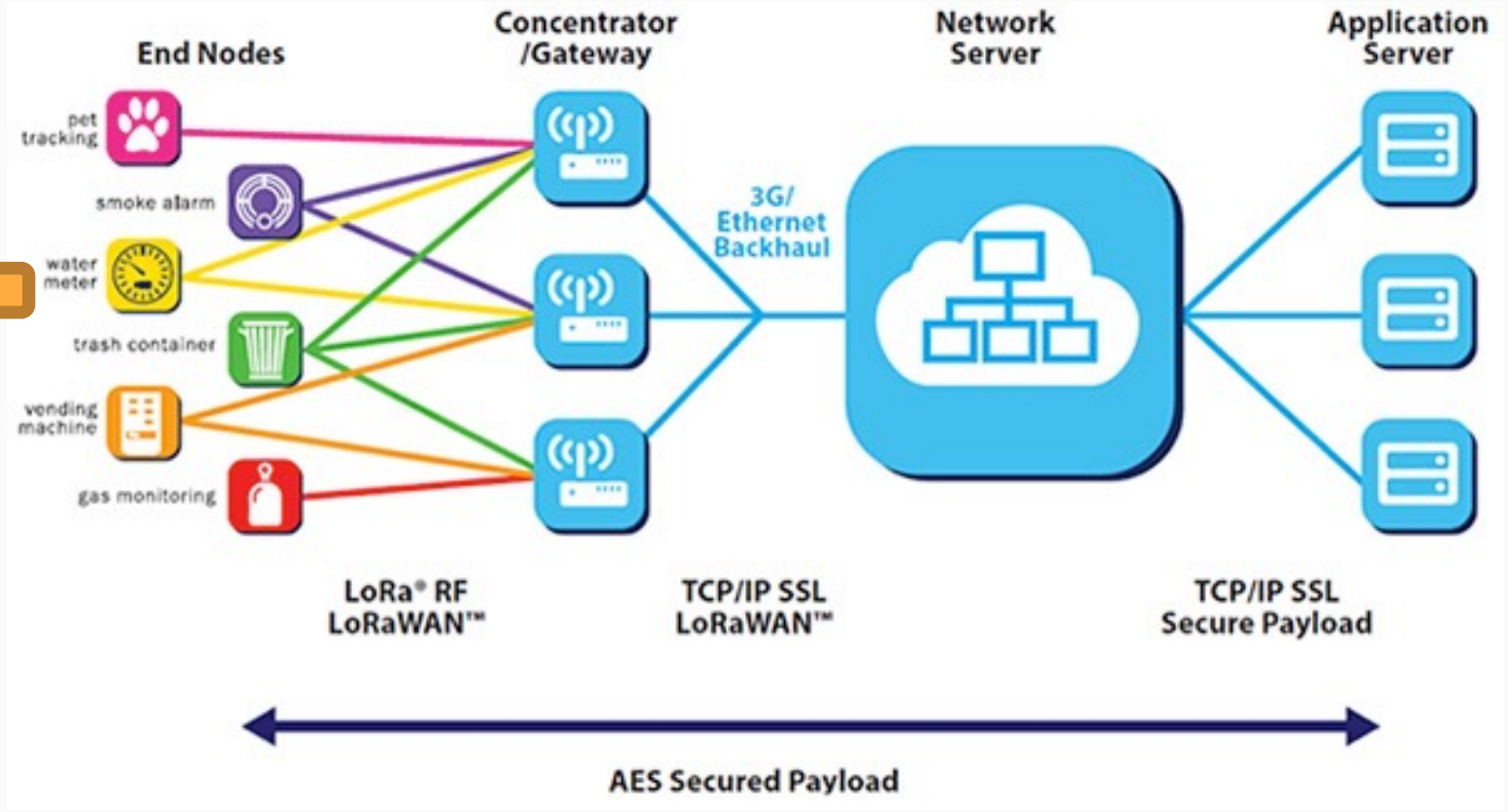
Data Rate	Spreading Factor	Bandwidth	Max Frame payload (Number N)
DR 0	SF12	125 kHz	51 bytes
DR 1	SF11	125 kHz	51 bytes
DR 2	SF10	125 kHz	51 bytes
DR 3	SF9	125 kHz	115 bytes
DR 4	SF8	125 kHz	242 bytes
DR 5	SF7	125 kHz	242 bytes
DR 6	SF7	250 kHz	242 bytes

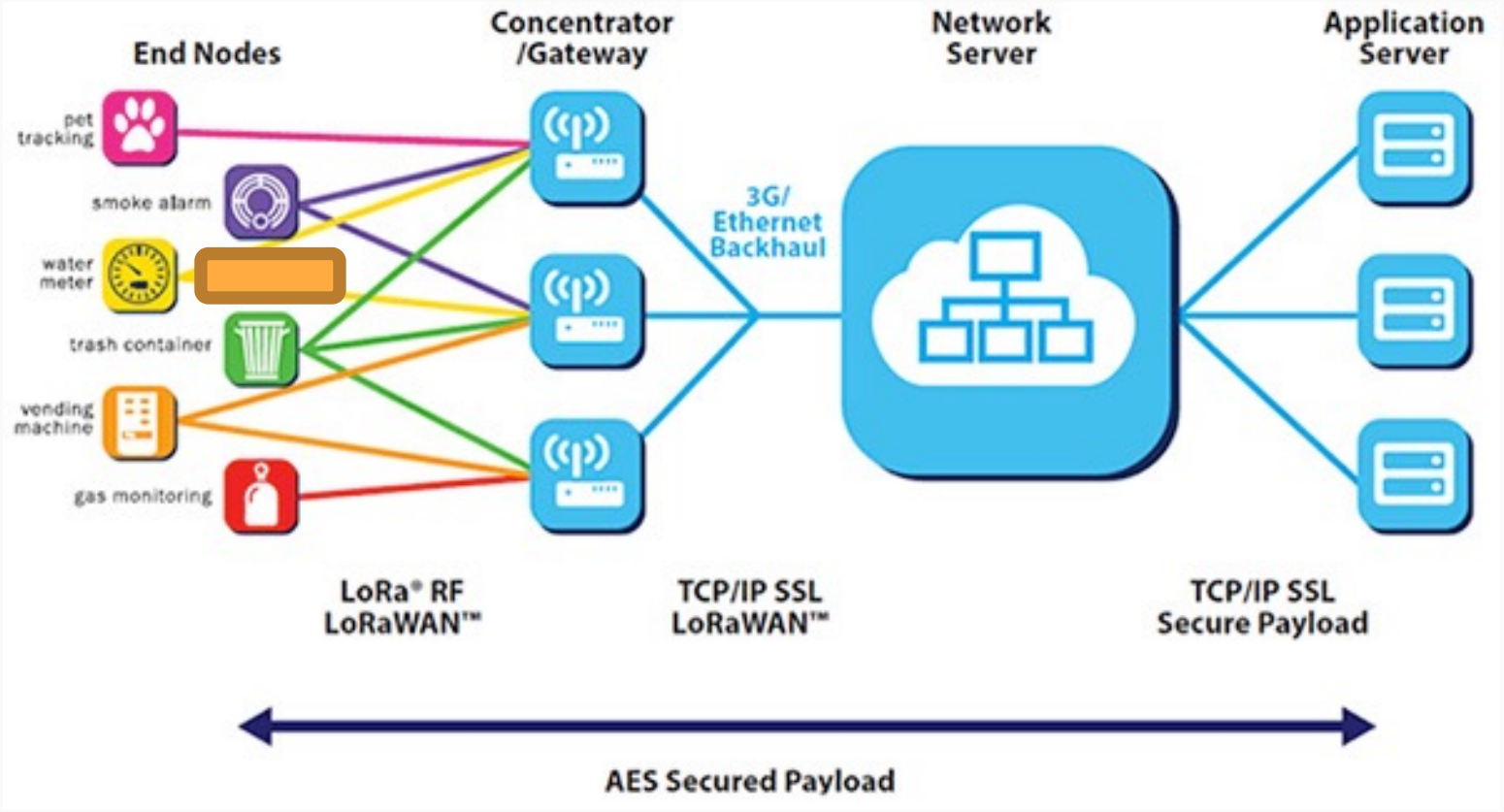
LoRaWAN characterizing parameters

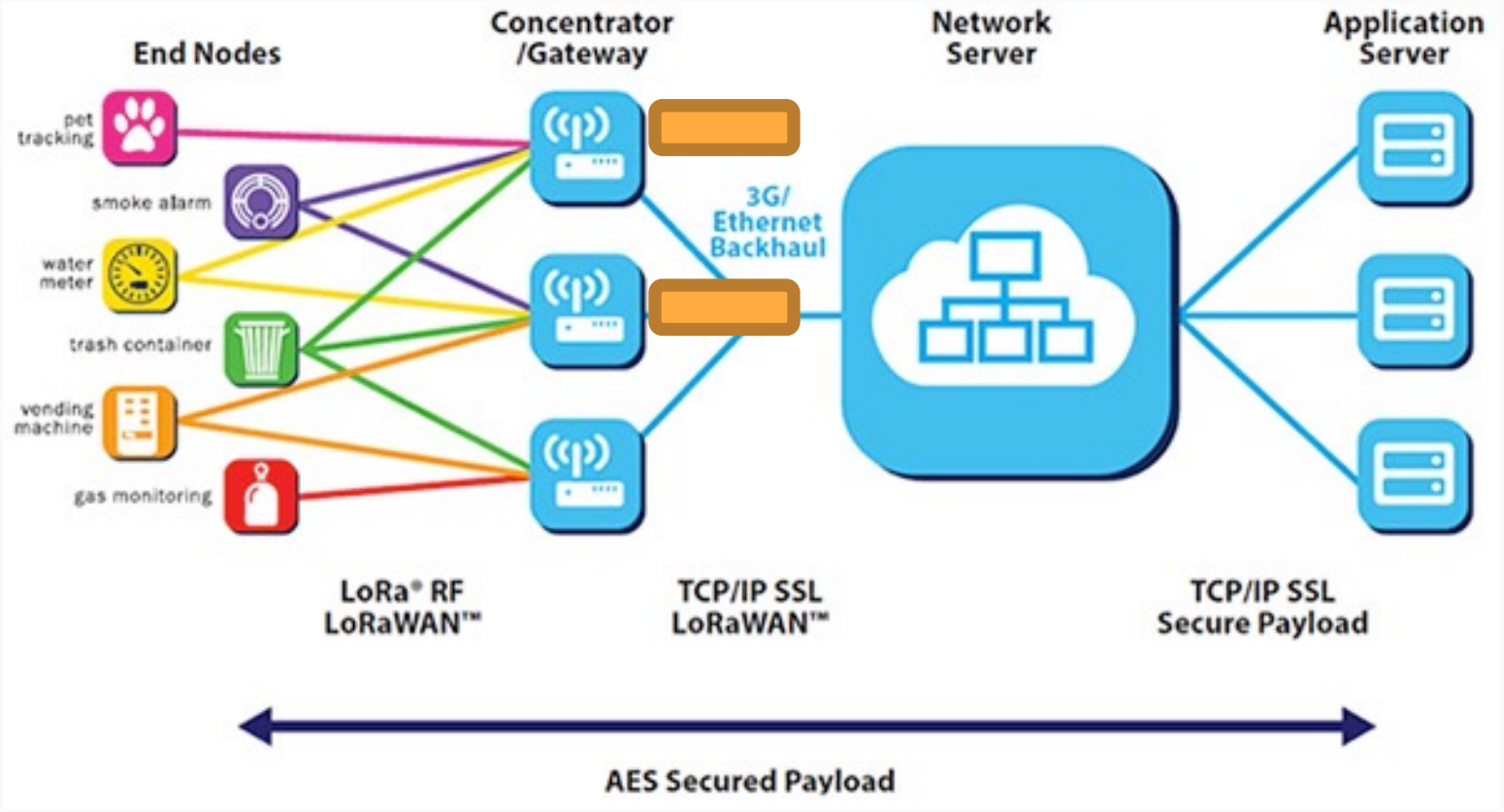


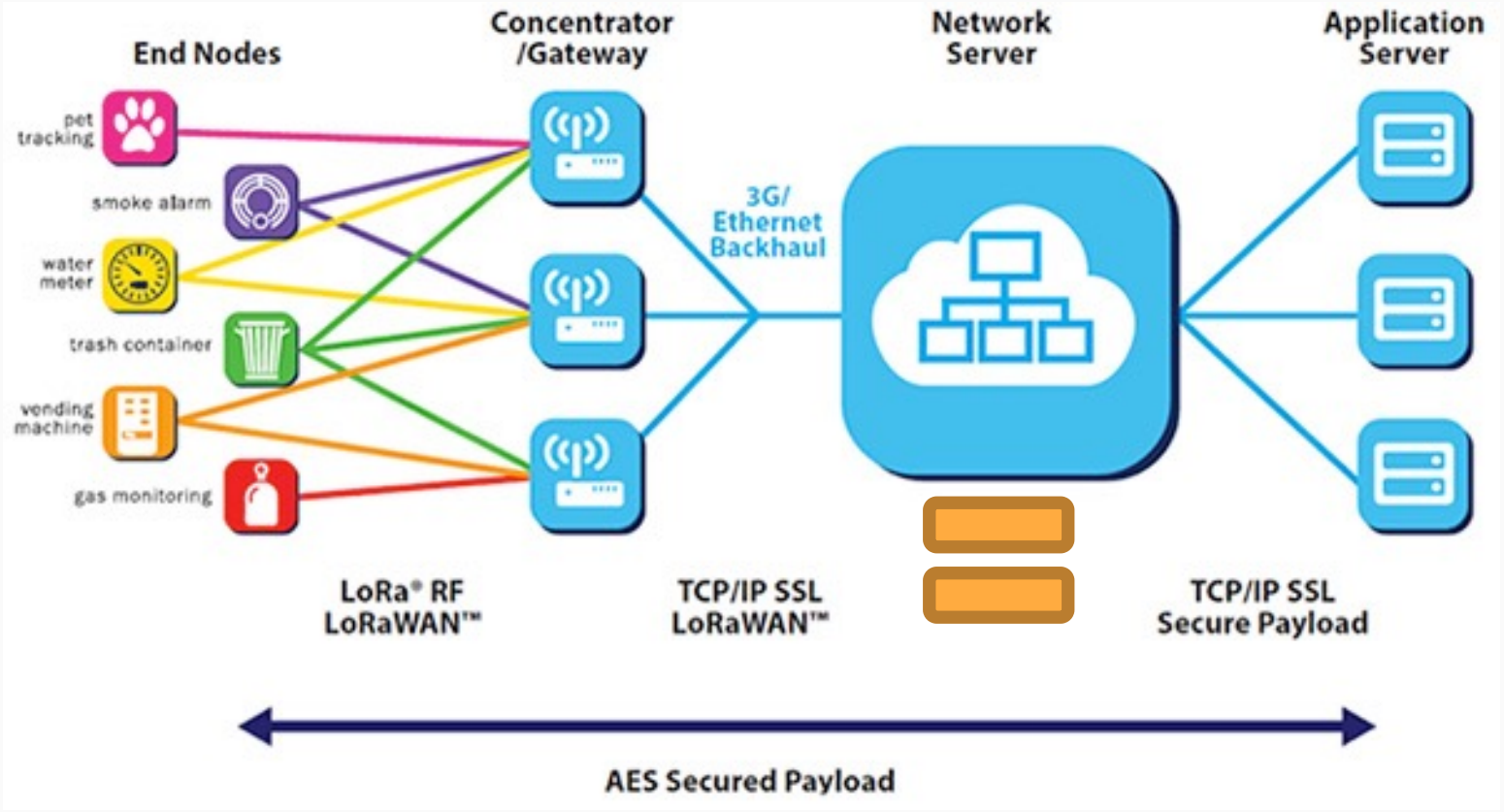

End Nodes
Concentrator /Gateway
Network Server
Application Server
pet tracking
smoke alarm
water meter
trash container
vending machine
gas monitoring
3G/ Ethernet Backhaul
LoRa® RF LoRaWAN™
TCP/IP SSL LoRaWAN™
TCP/IP SSL Secure Payload
AES Secured Payload

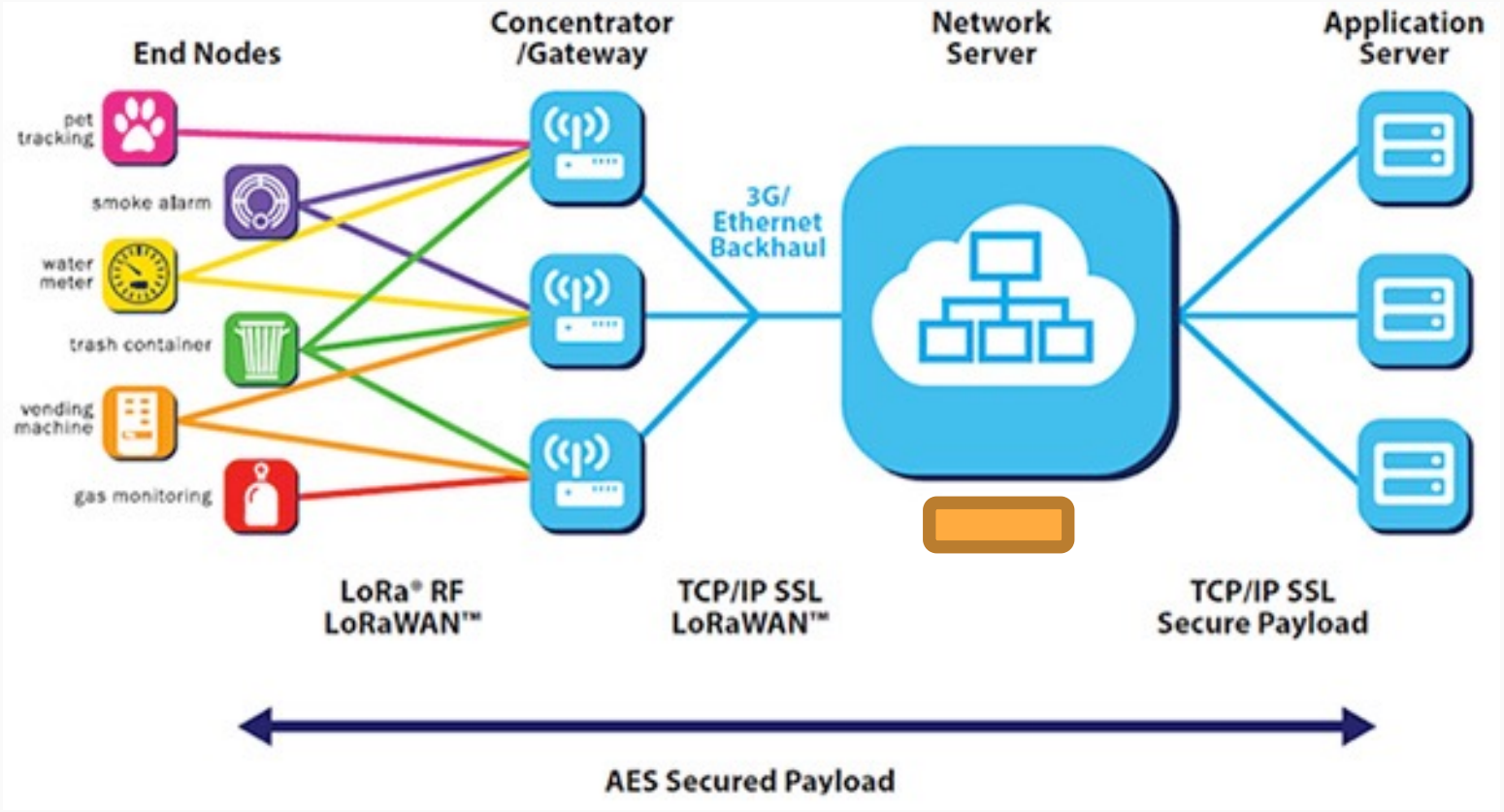
123

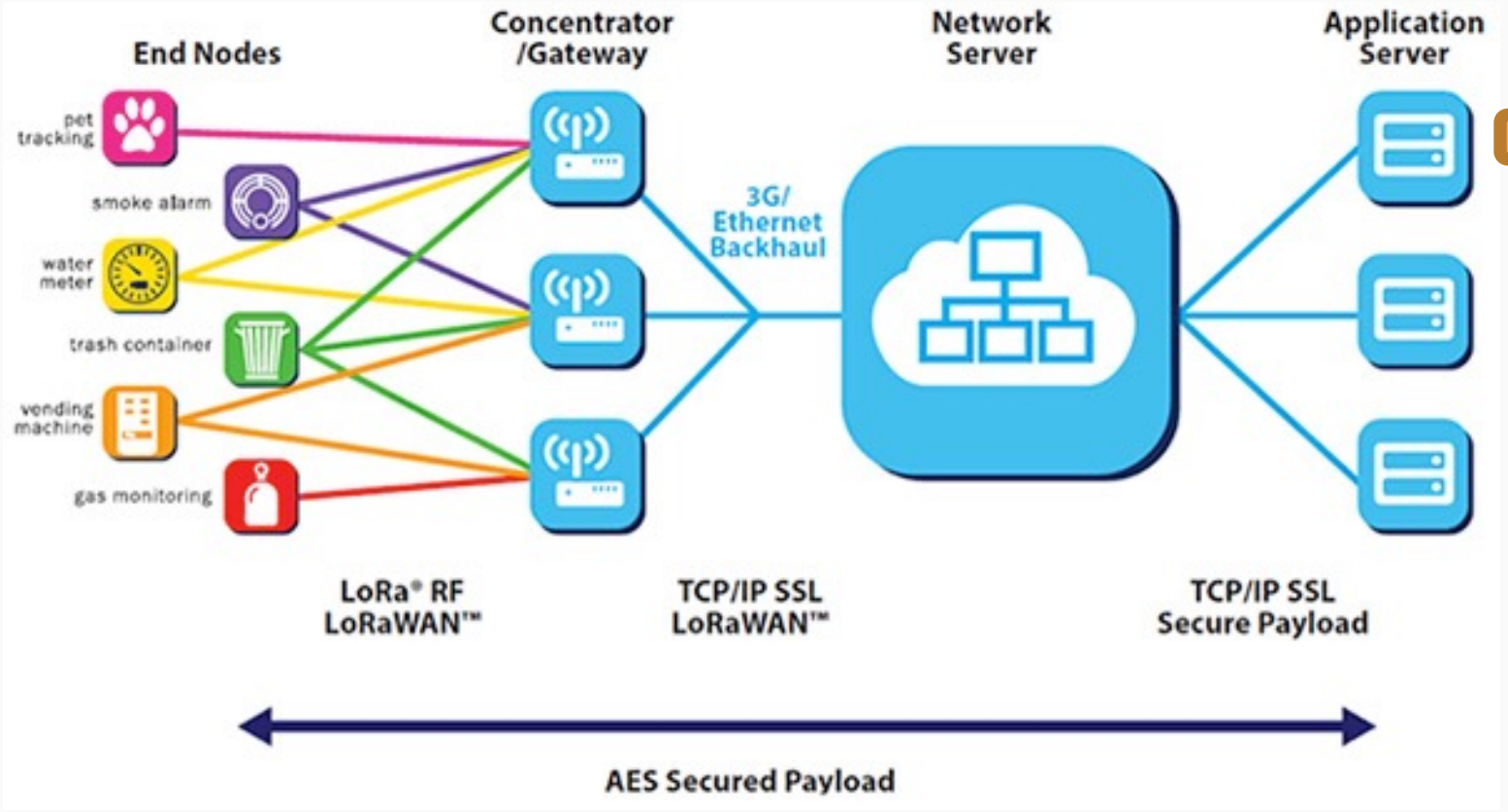

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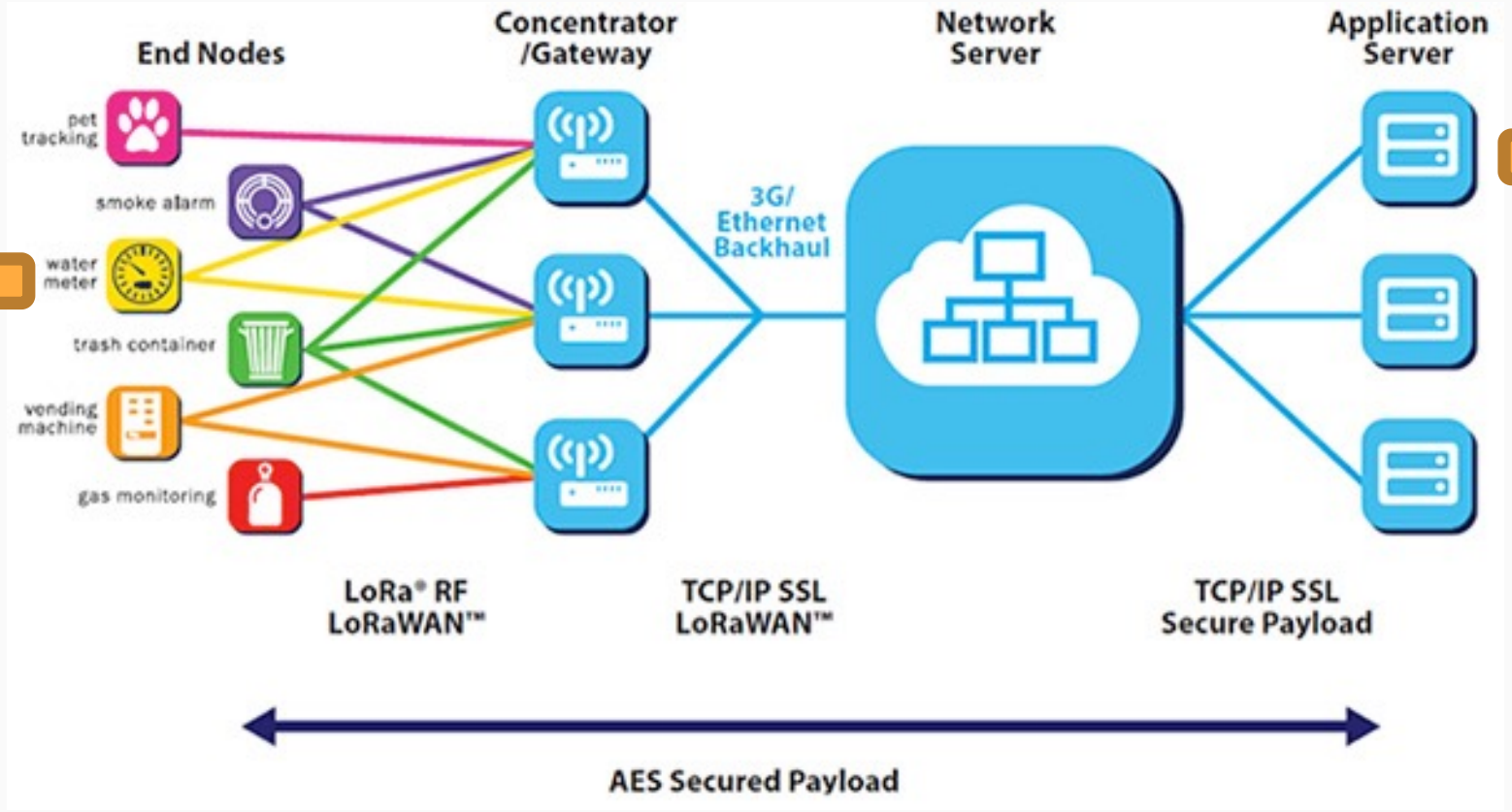

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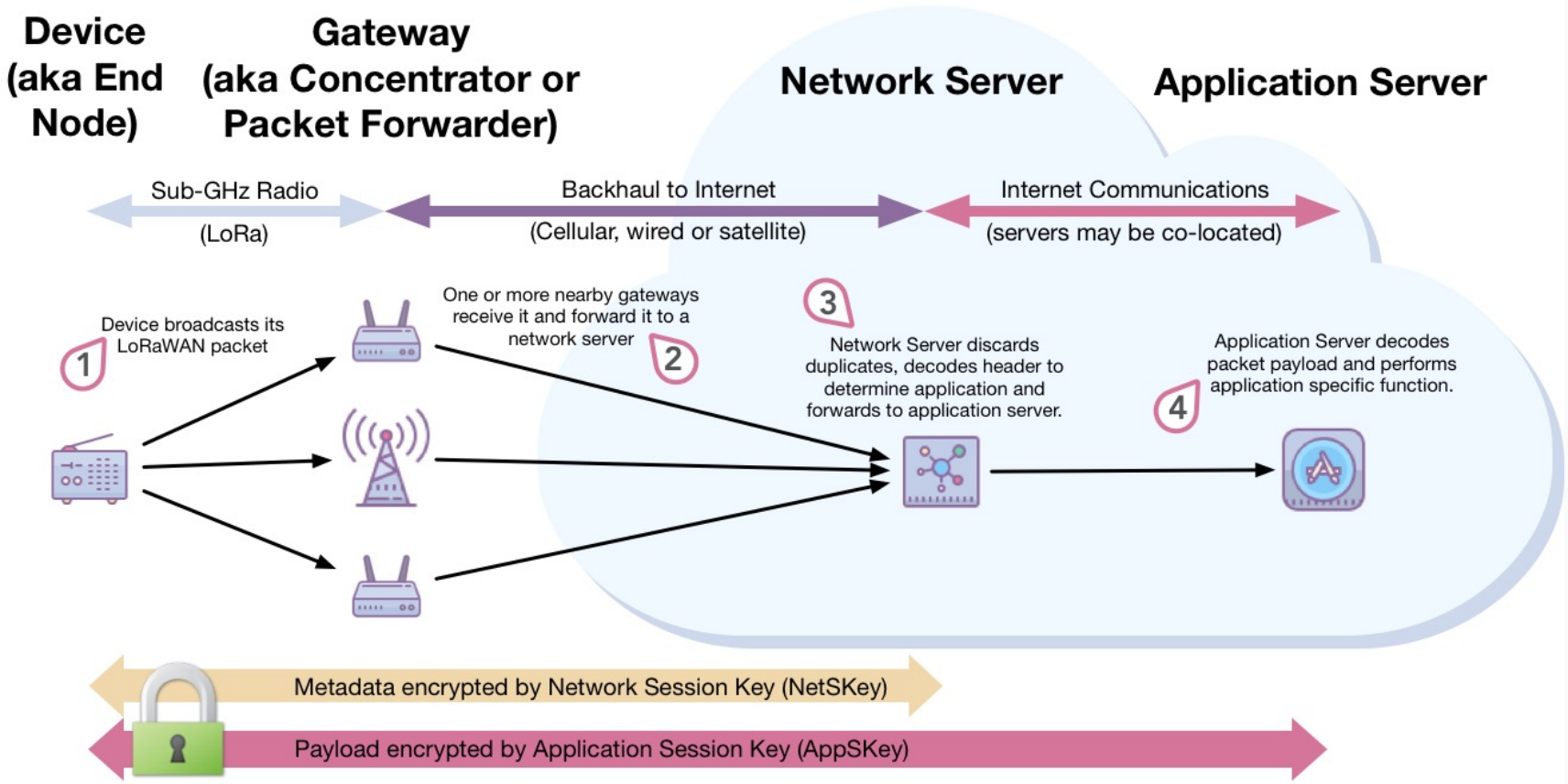

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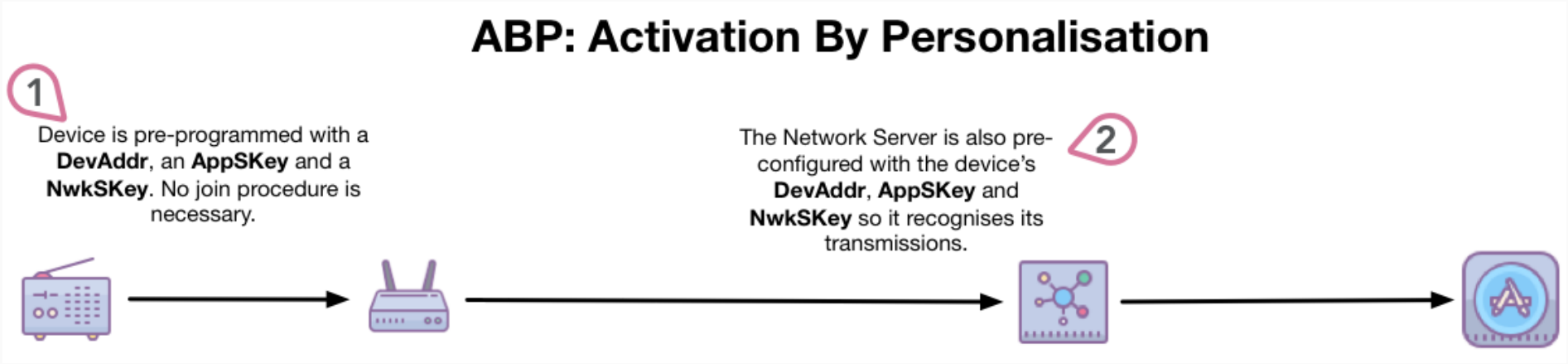

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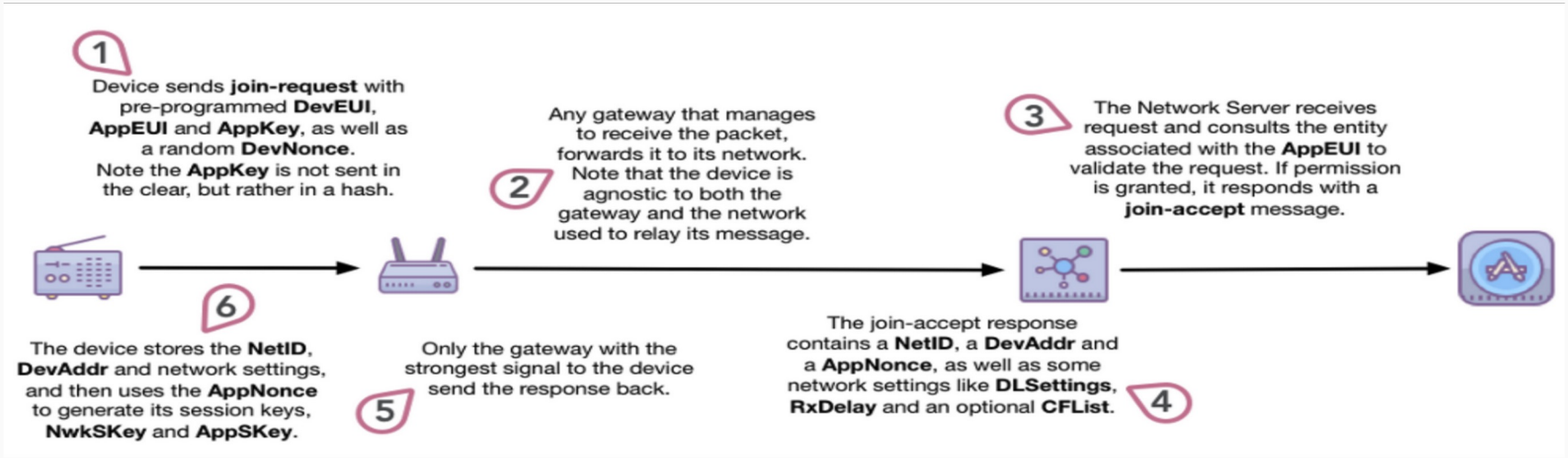
LoRaWAN architecture



LoRaWAN architecture



OTAA: Devices activation

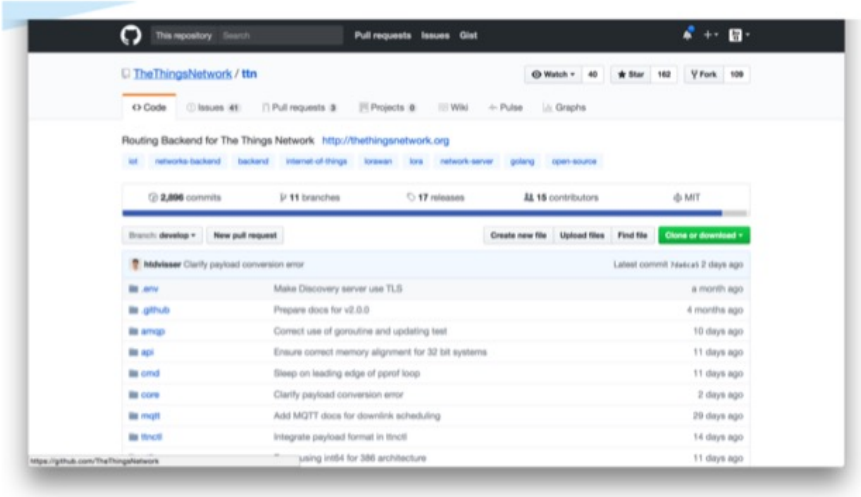


The Things Network Manifesto

Everything that carries power will be connected to Internet eventually.

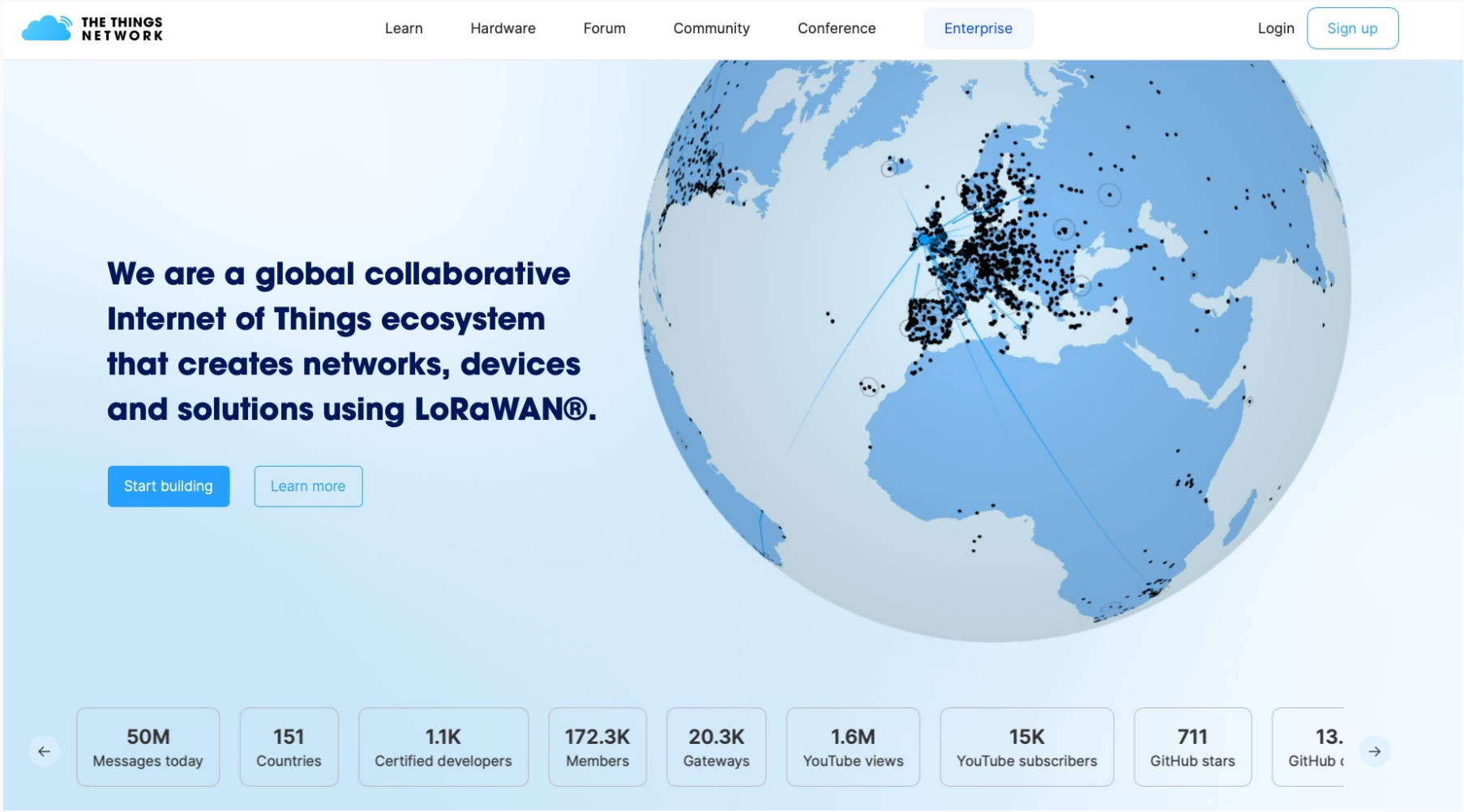
Controlling the network that makes this possible means controlling the world. ***We believe that this power should not be restricted to a few people, companies or nations. Instead this should be distributed over as many people as possible without the possibility to be taken away by anyone.*** We therefore founded "The Things Network".

TTN

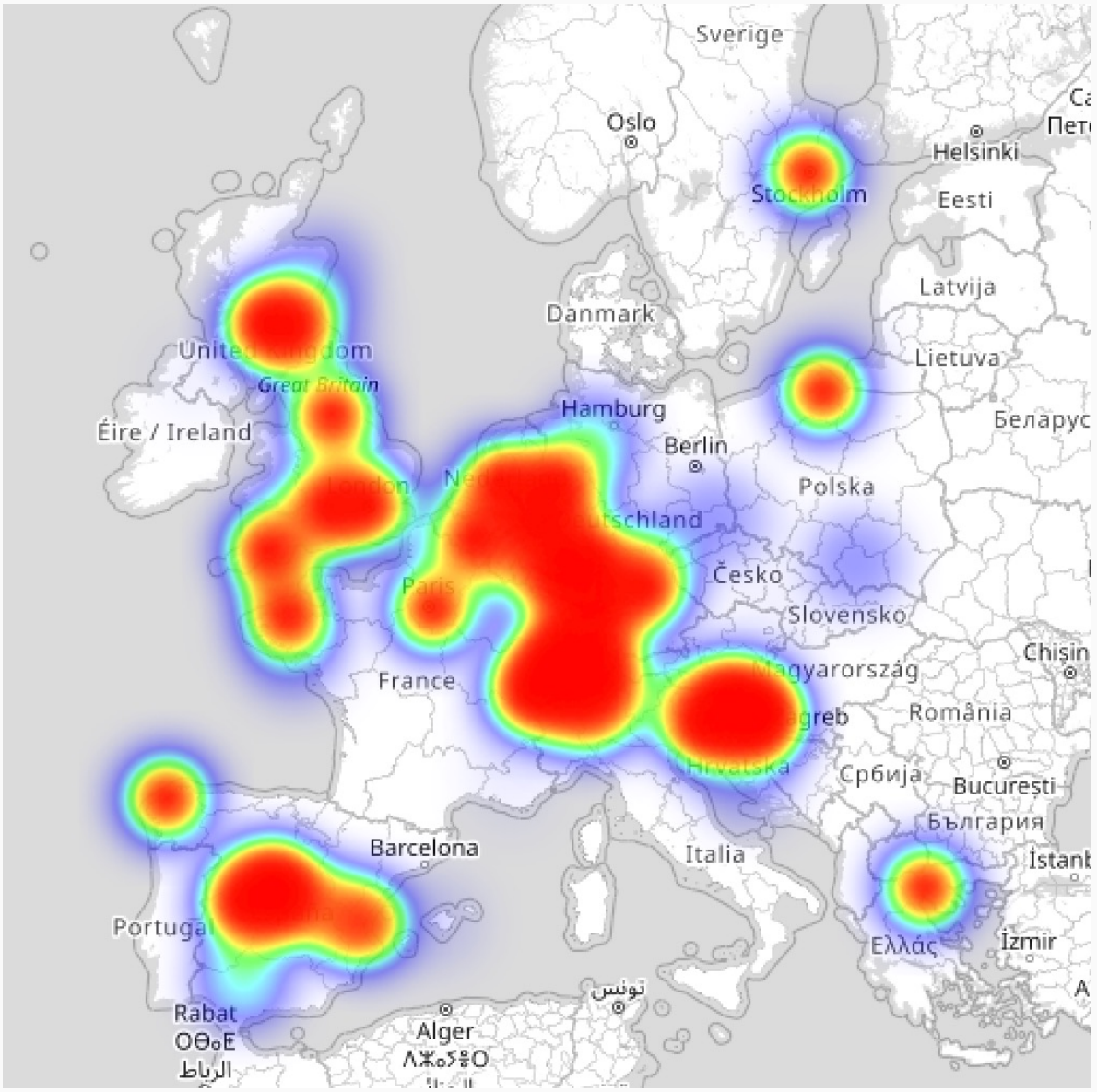


OPEN SOURCE BACKEND

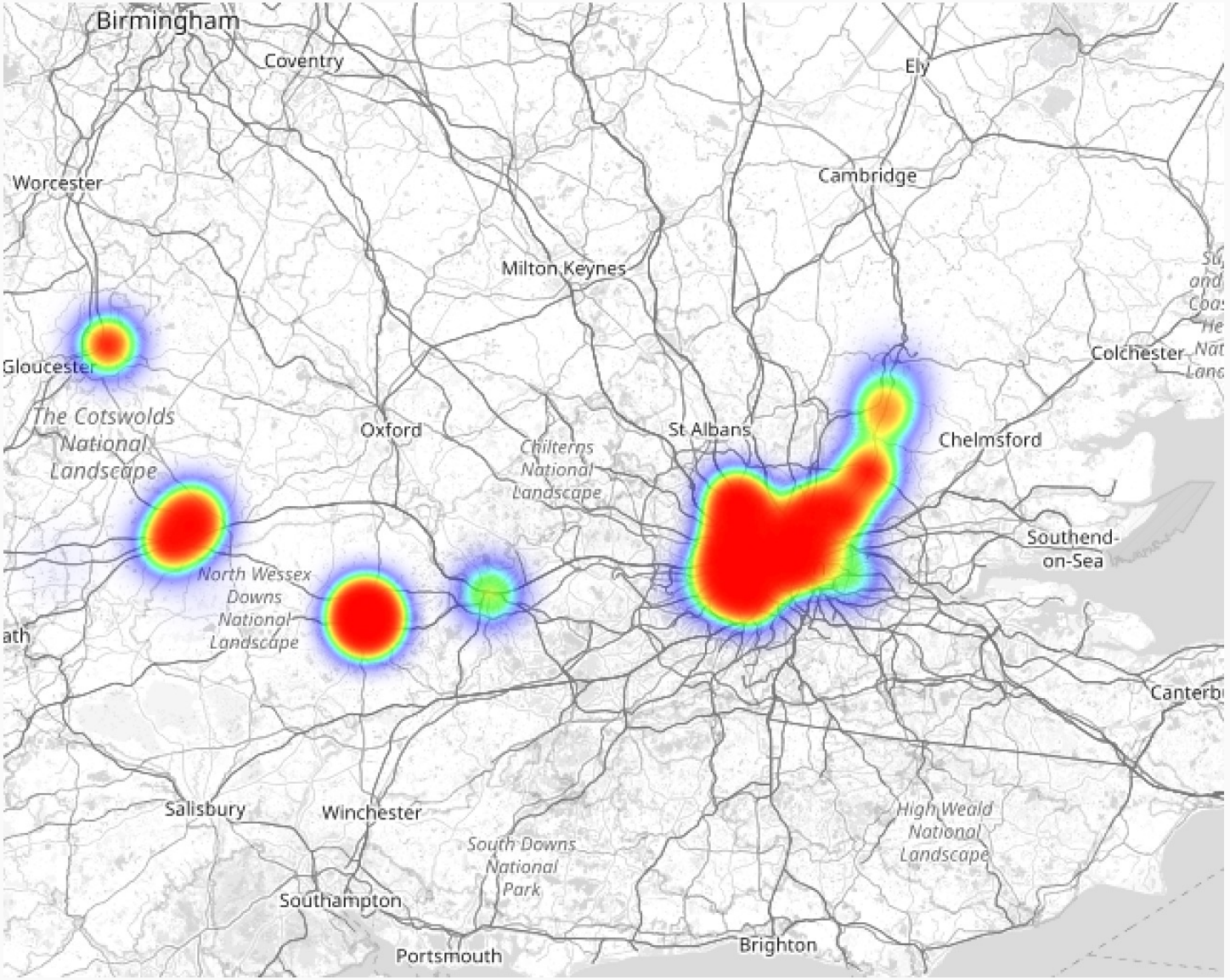
TTN


THE THINGS NETWORK
Learn
Hardware
Forum
Community
Conference
Enterprise
Login
Sign up
We are a global collaborative Internet of Things ecosystem that creates networks, devices and solutions using LoRaWAN®.
Start building
Learn more
50M
Messages today
151
Countries
1.1K
Certified developers
172.3K
Members
20.3K
Gateways
1.6M
YouTube views
15K
YouTube subscribers
711
GitHub stars
13.
GitHub

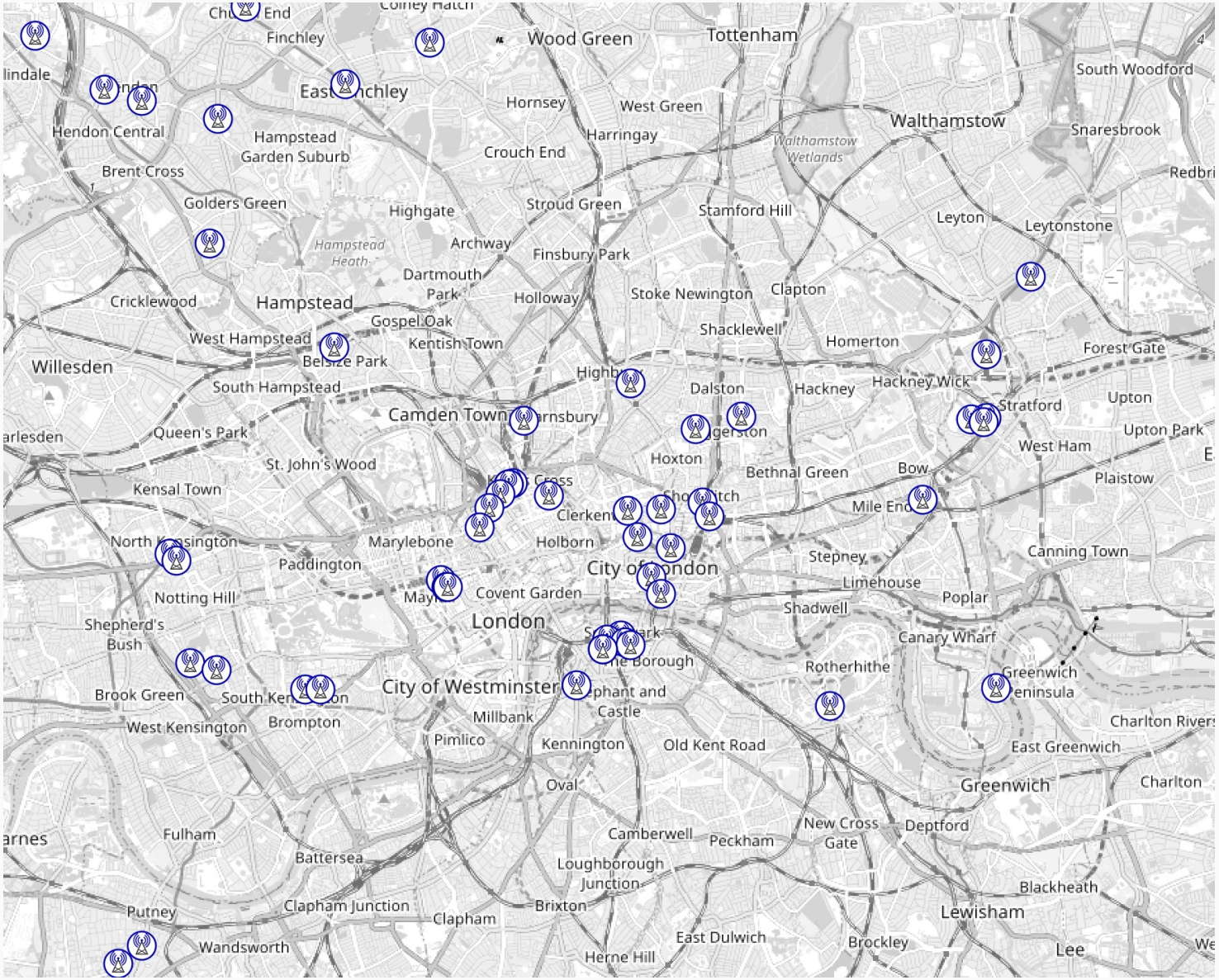
TTN


Sverige
Oslo
Helsinki
Stockholm
Eesti
Latvija
Danmark
Lietuva
Беларус
Hamburg
Berlin
Polska
Česko
Slovensko
Magyarország
Chișin
România
București
България
Србија
Hrvatska
France
Paris
Barcelona
Italia
İstanb
İzmir
Ελλάς
Portuga
Rabat
Alger
تونس
الرباط

TTN


Birmingham
Coventry
Ely
Worcester
Cambridge
Milton Keynes
Gloucester
Colchester
The Cotswolds National Landscape
Oxford
Chilterns National Landscape
St Albans
Chelmsford
Southend-on-Sea
North Wessex Downs National Landscape
Salisbury
Winchester
South Downs National Park
High Weald National Landscape
Southampton
Portsmouth
Brighton

TTN

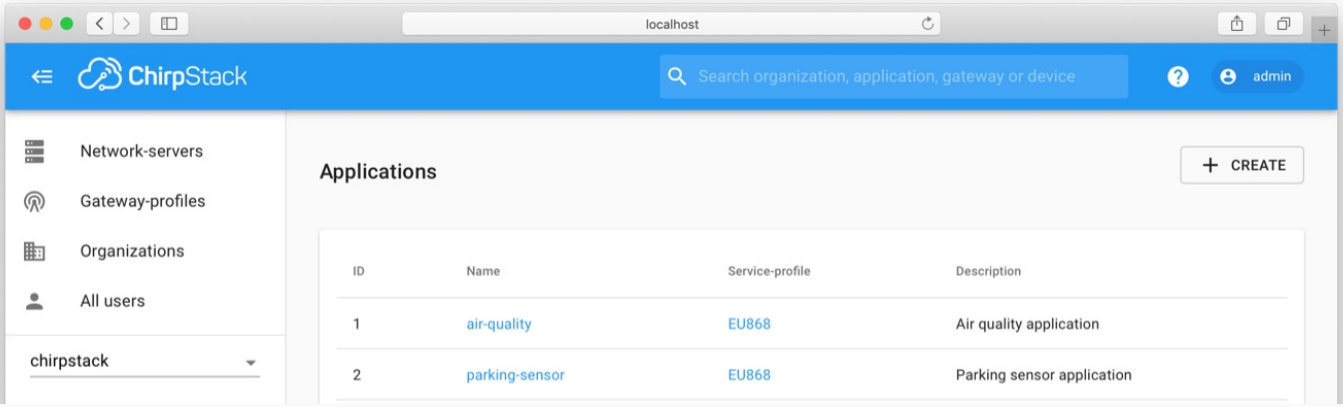

Finchley
Wood Green
Tottenham
South Woodford
Hendon Central
Hampstead Garden Suburb
Hornsey
West Green
Harringay
Walthamstow
Walthamstow Wetlands
Snaresbrook
Brent Cross
Crouch End
Golders Green
Highgate
Stroud Green
Stamford Hill
Leyton
Leytonstone
Hampstead Heath
Archway
Finsbury Park
Dartmouth Park
Holloway
Stoke Newington
Clapton
Cricklewood
Hampstead
Gospel Oak
Shacklewell
West Hampstead
Kentish Town
Homerton
Forest Gate
Willesden
Belsize Park
Hackney
Hackney Wick
South Hampstead
Dalston
Stratford
Upton
Camden Town
Queen's Park
Upton Park
West Ham
St. John's Wood
Hoxton
Bethnal Green
Bow
Plaistow
Kensal Town
Clerkenwell
Mile End
North Kensington
Marylebone
Holborn
Paddington
City of London
Stepney
Canning Town
Limehouse
Notting Hill
Covent Garden
Poplar
Shadwell
London
Shepherd's Bush
Canary Wharf
Rotherhithe
Greenwich Peninsula
Brook Green
City of Westminster
Elephant and Castle
Millbank
Brompton
West Kensington
Pimlico
Kennington
Old Kent Road
East Greenwich
Charlton
Oval
Greenwich
Fulham
Camberwell
Peckham
New Cross Gate
Deptford
Battersea
Loughborough Junction
Blackheath
Putney
Clapham Junction
Brixton
Clapham
Lewisham
Wandsworth
East Dulwich
Brockley
Herne Hill
Lee

Chirpstack

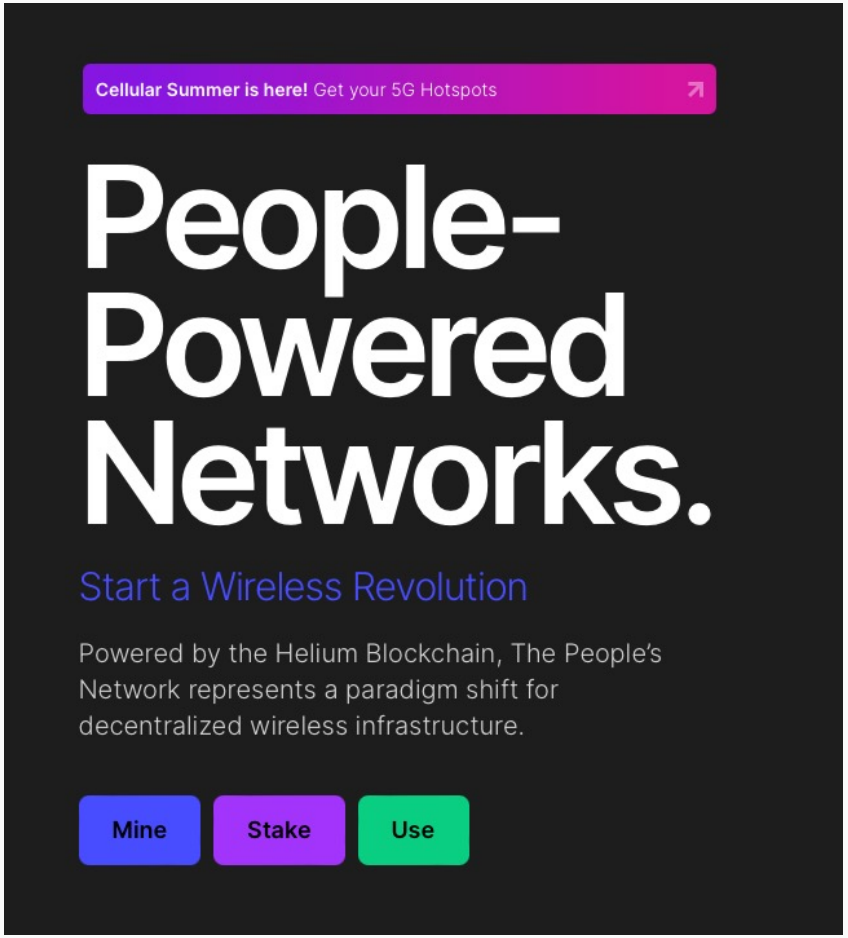
ChirpStack, open-source LoRaWAN® Network Server stack

The ChirpStack open-source LoRaWAN Network Server stack provides open-source components for LoRaWAN networks. Together they form a ready-to-use solution including an user-friendly web-interface for device management and APIs for integration. The modular architecture makes it possible to integrate within existing infrastructures. All components are licensed under the MIT license and can be used for commercial purposes.

Get started



Helium



~21,000

The Things Network (TTN) gateways worldwide

~370,000

Helium IoT hotspots online (peak ~990,000 in 2022)

Helium has roughly 18× more gateways than TTN

Approximate figures, 2024. Source: The Things Network website; Helium Explorer.

Helium


HARLOW
ST ALBANS
CHIPPING ONG
WALTHAM ABBEY
WATFORD
CHIPPING BARNET
COLLIER ROW
HARROW
WALTHAMSTOW
DAGENHAM
SOUTHALL
LONDON
BEXLEYHEATH
CATFORD
CHERTSEY
SURBITON
ORPINGTON
CROYDON
COBHAM
EPSOM

LoRaWAN products



LoRa Alliance: LoRaWAN Certified Devices

Official list of end devices that passed LoRaWAN certification

lora-alliance.org/lorawan-certified-devices



The Things Network Device Repository

Searchable catalogue of devices with specs, photos and payload decoders

thethingsnetwork.org/device-repository



TTN lorawan-devices on GitHub

Open-source data behind the repository; vendors add devices here

github.com/TheThingsNetwork/lorawan-devices

LoRaWAN tracking devices



Gateways: examples

INDOOR

~€70

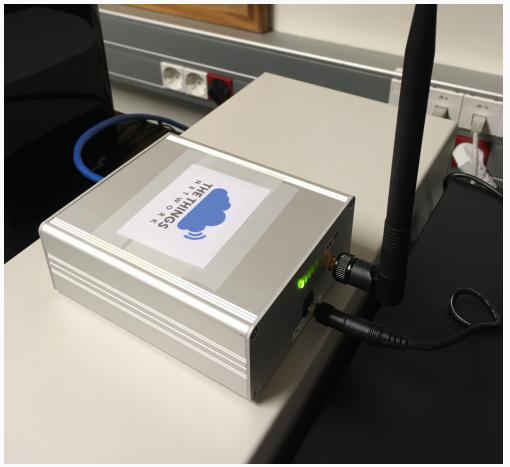
e.g. The Things Indoor Gateway: Wi-Fi backhaul, plugs into a wall socket

OUTDOOR

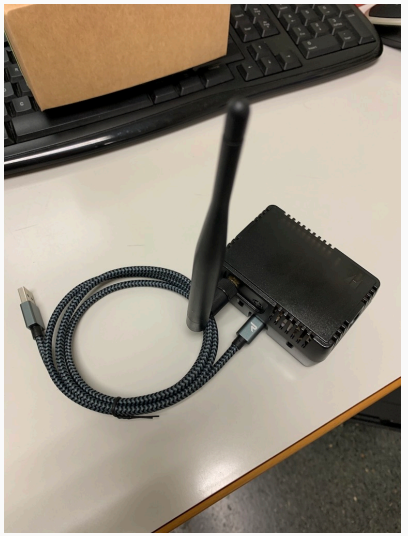
€500–1,500

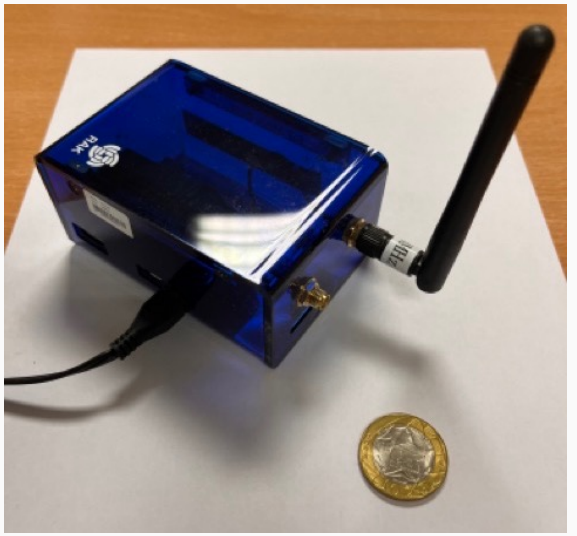
IP67 enclosure, external antenna, PoE or cellular backhaul

An outdoor gateway costs 7–20× an indoor one







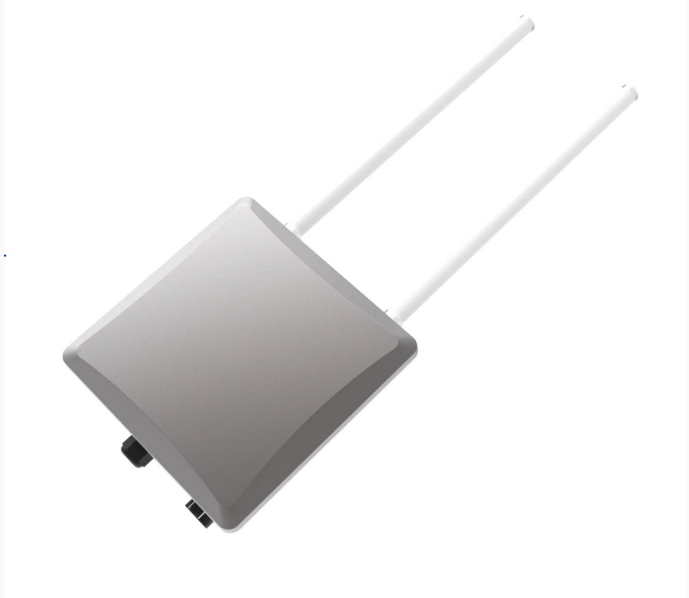


Indicative list prices, 2024, excl. VAT and installation.

Gateways: examples







Main takeaways

1 **IoT can trade speed for reach**

Most sensors tolerate low throughput, delays and long sleep, which is what LPWAN gives up to gain range and battery life.

2 **The link budget decides range**

Transmit power plus antenna gains minus path loss must beat receiver sensitivity; keep the first Fresnel zone clear.

3 **LoRa: chirps in sub-GHz ISM bands**

Chirp spread spectrum at 433/868/915 MHz, license-free but regulated: 1% duty cycle in the EU (36 s per hour).

4 **Spreading factor is the main trade-off**

SF7 sends 6,836 bps; SF12 only 366 bps, but reaches further at the cost of airtime and battery.

5 **LoRaWAN is the open network layer**

Star-of-stars topology: devices talk only to gateways, in classes A/B/C, with ADR and built-in encryption.

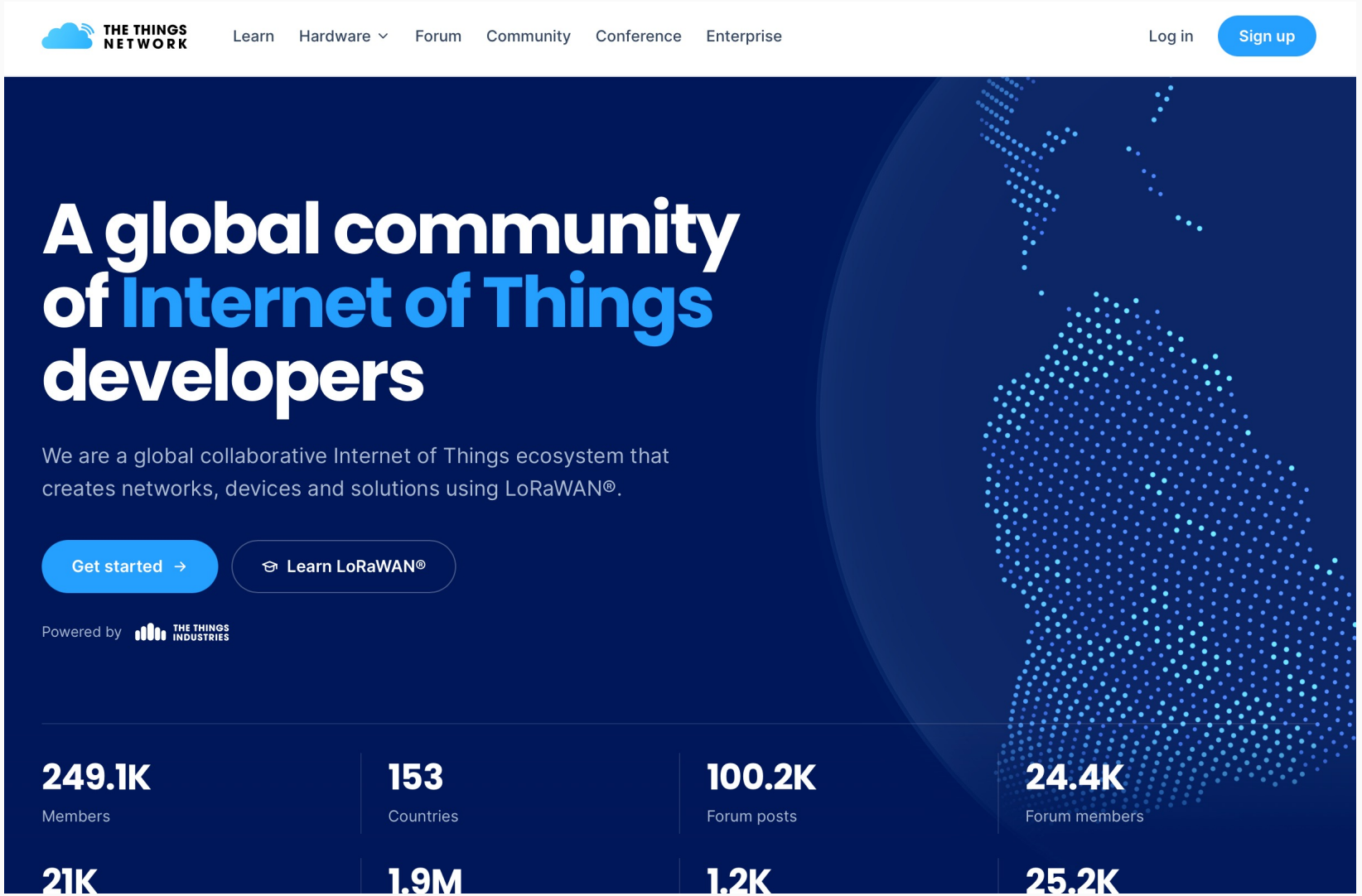
6 **Open networks, affordable gateways**

TTN has ~21,000 gateways, Helium ~370,000 hotspots; an indoor gateway costs ~€70, outdoor €500–1,500.

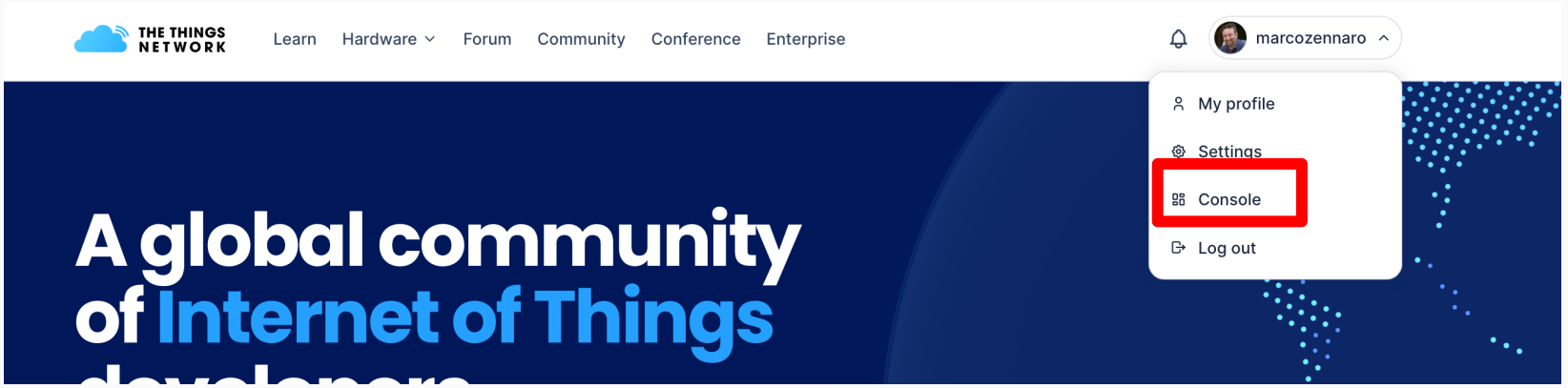
Lab time!

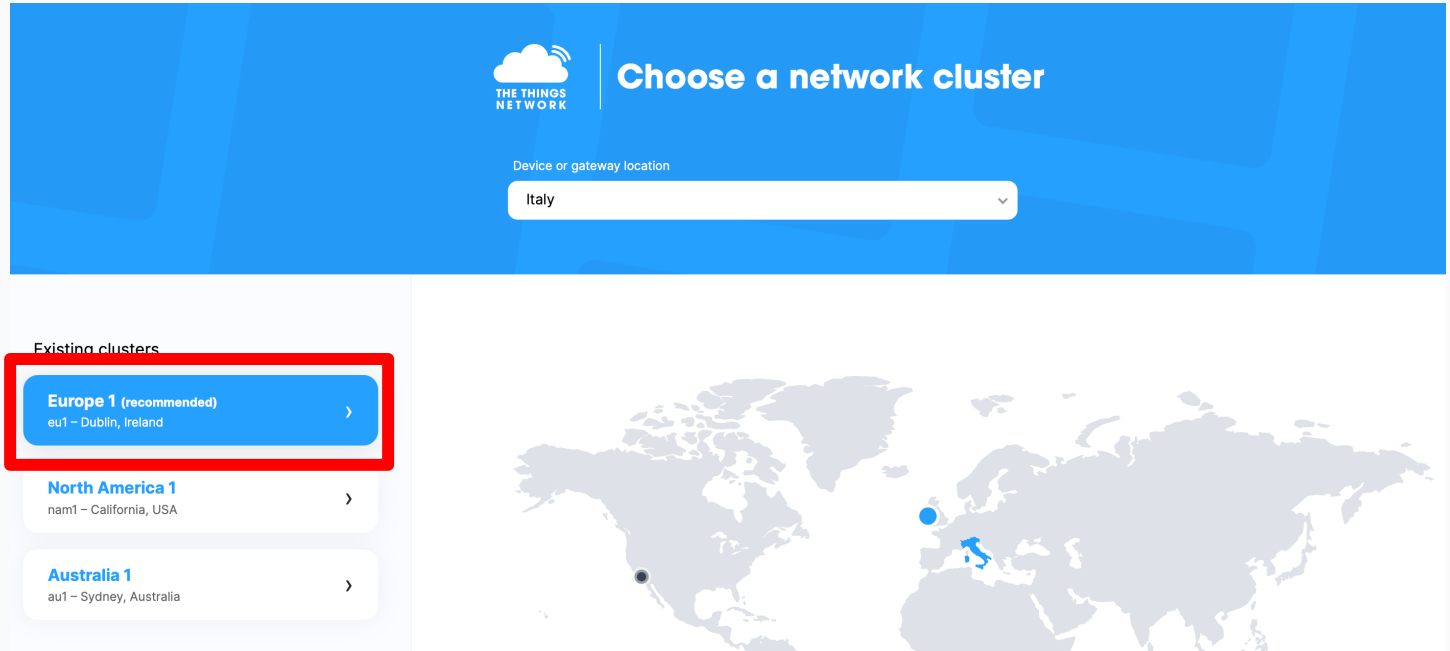
Lab

https://www.thethingsnetwork.org

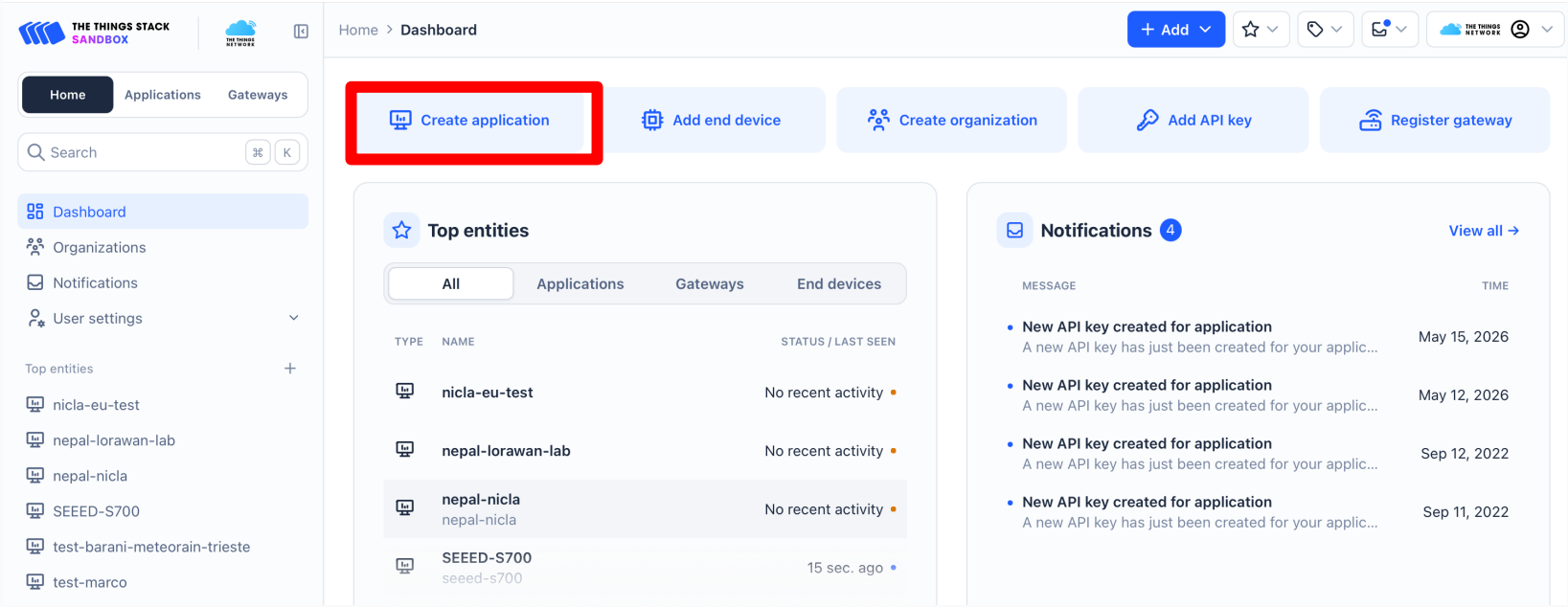


Lab

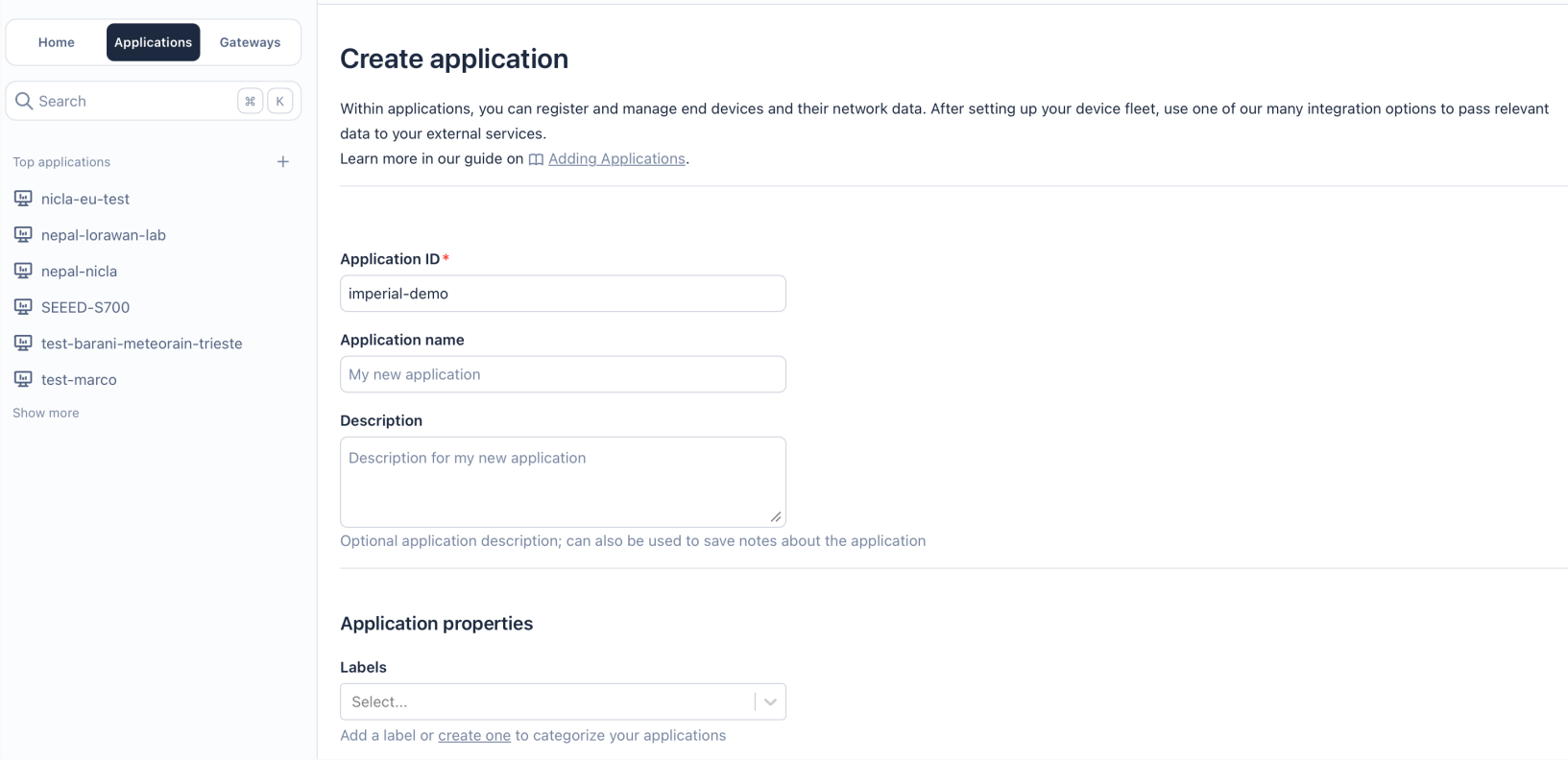

THE THINGS NETWORK
Learn
Hardware
Forum
Community
Conference
Enterprise
marcozennaro
My profile
Settings
Console
Log out
A global community of Internet of Things


THE THINGS NETWORK
Choose a network cluster
Device or gateway location
Italy
Existing clusters
Europe 1 (recommended)
eu1 – Dublin, Ireland
North America 1
nam1 – California, USA
Australia 1
au1 – Sydney, Australia

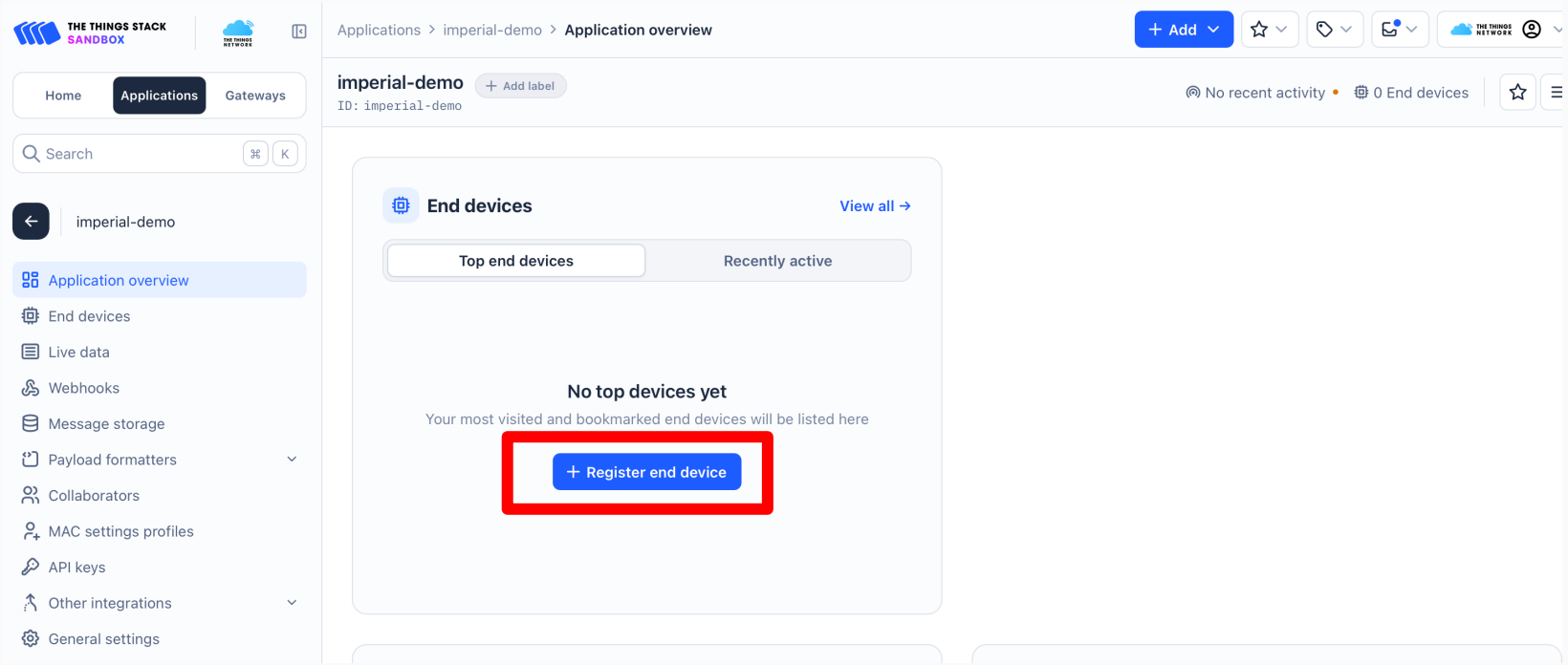
Lab


THE THINGS STACK SANDBOX
Home
Applications
Gateways
Search
Dashboard
Organizations
Notifications
User settings
Top entities
nicla-eu-test
nepal-lorawan-lab
nepal-nicla
SEEED-S700
test-barani-meteorain-trieste
test-marco
Home > Dashboard
+ Add
Create application
Add end device
Create organization
Add API key
Register gateway
Top entities
All
Applications
Gateways
End devices
TYPE
NAME
STATUS / LAST SEEN
nicla-eu-test
No recent activity
nepal-lorawan-lab
No recent activity
nepal-nicla
nepal-nicla
No recent activity
SEEED-S700
seeed-s700
15 sec. ago
Notifications 4
View all →
MESSAGE
TIME
New API key created for application
A new API key has just been created for your applic...
May 15, 2026
New API key created for application
A new API key has just been created for your applic...
May 12, 2026
New API key created for application
A new API key has just been created for your applic...
Sep 12, 2022
New API key created for application
A new API key has just been created for your applic...
Sep 11, 2022

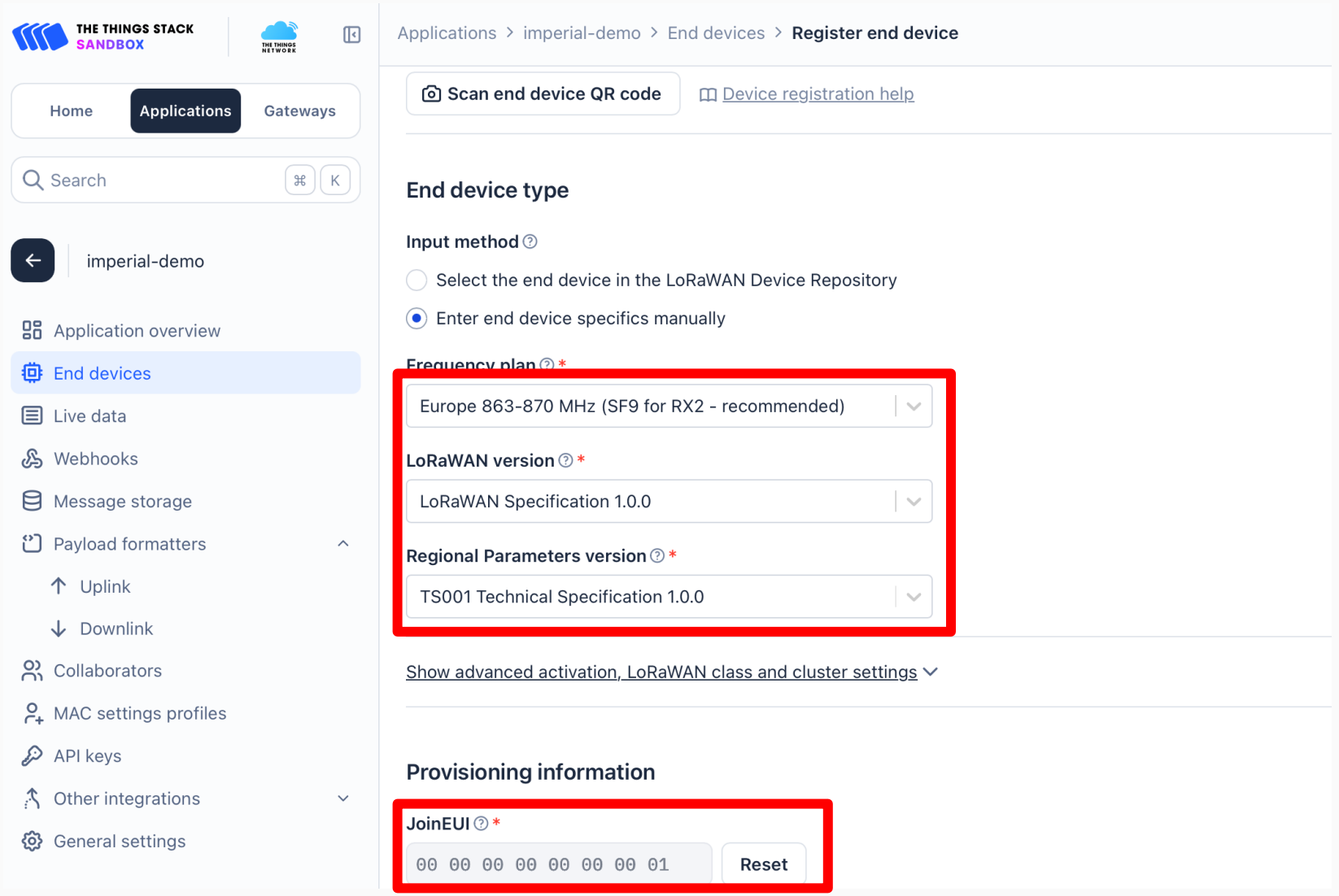
Lab


Home
Applications
Gateways
Search
⌘
K
Top applications
nicla-eu-test
nepal-lorawan-lab
nepal-nicla
SEEED-S700
test-barani-meteorain-trieste
test-marco
Show more
Create application
Within applications, you can register and manage end devices and their network data. After setting up your device fleet, use one of our many integration options to pass relevant data to your external services.
Learn more in our guide on Adding Applications.
Application ID *
imperial-demo
Application name
My new application
Description
Description for my new application
Optional application description; can also be used to save notes about the application
Application properties
Labels
Select...
Add a label or create one to categorize your applications

Lab

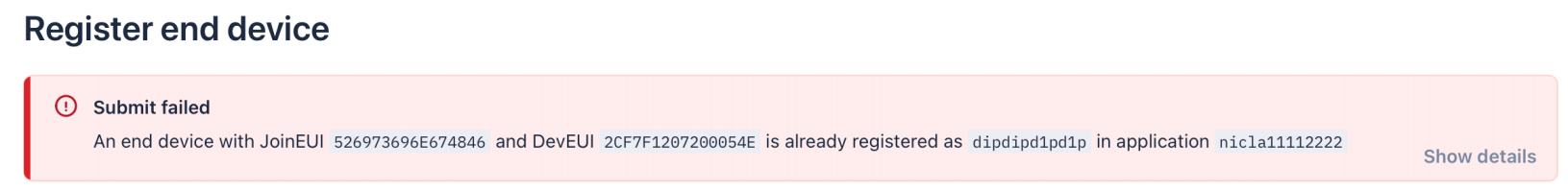

THE THINGS STACK SANDBOX
Home
Applications
Gateways
Search
imperial-demo
Application overview
End devices
Live data
Webhooks
Message storage
Payload formatters
Collaborators
MAC settings profiles
API keys
Other integrations
General settings
Applications > imperial-demo > Application overview
+ Add
imperial-demo
+ Add label
ID: imperial-demo
No recent activity
0 End devices
End devices
View all →
Top end devices
Recently active
No top devices yet
Your most visited and bookmarked end devices will be listed here
+ Register end device

Lab


THE THINGS STACK SANDBOX
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imperial-demo
Application overview
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Webhooks
Message storage
Payload formatters
Uplink
Downlink
Collaborators
MAC settings profiles
API keys
Other integrations
General settings
Applications > imperial-demo > End devices > Register end device
Scan end device QR code
Device registration help
End device type
Input method
Select the end device in the LoRaWAN Device Repository
Enter end device specifics manually
Frequency plan *
Europe 863-870 MHz (SF9 for RX2 - recommended)
LoRaWAN version *
LoRaWAN Specification 1.0.0
Regional Parameters version *
TS001 Technical Specification 1.0.0
Show advanced activation, LoRaWAN class and cluster settings
Provisioning information
JoinEUI *
00 00 00 00 00 00 00 01
Reset

Orphaned devices

Issue:



If a device has been already registered in a TTN application, it cannot be registered in another application unless it is first deleted from the original application.

Solution: give the device a random JoinEUI value (0000000000000001 or 0000000003400001 or any other random value)

LoRaWAN join values

1



AppEUI

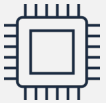
also called JoinEUI

You choose it

Any 16-digit hex value works.

`0000000000000001`

2



DevEUI

unique device ID

Read from the device

Fixed in the LoRa module. Read it over serial:

`AT+ID=DevEui`

3



AppKey

secret join key

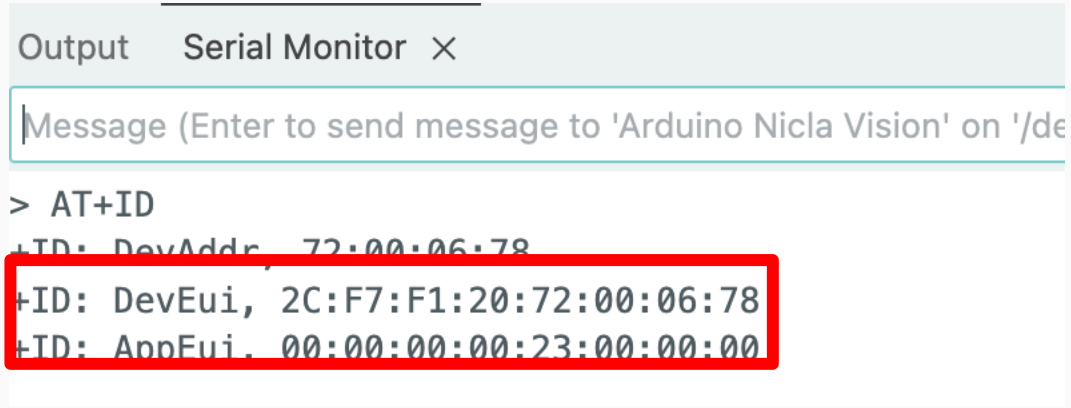
Generated by TTN

TTN creates it after you enter AppEUI and DevEUI. Copy it into the sketch.

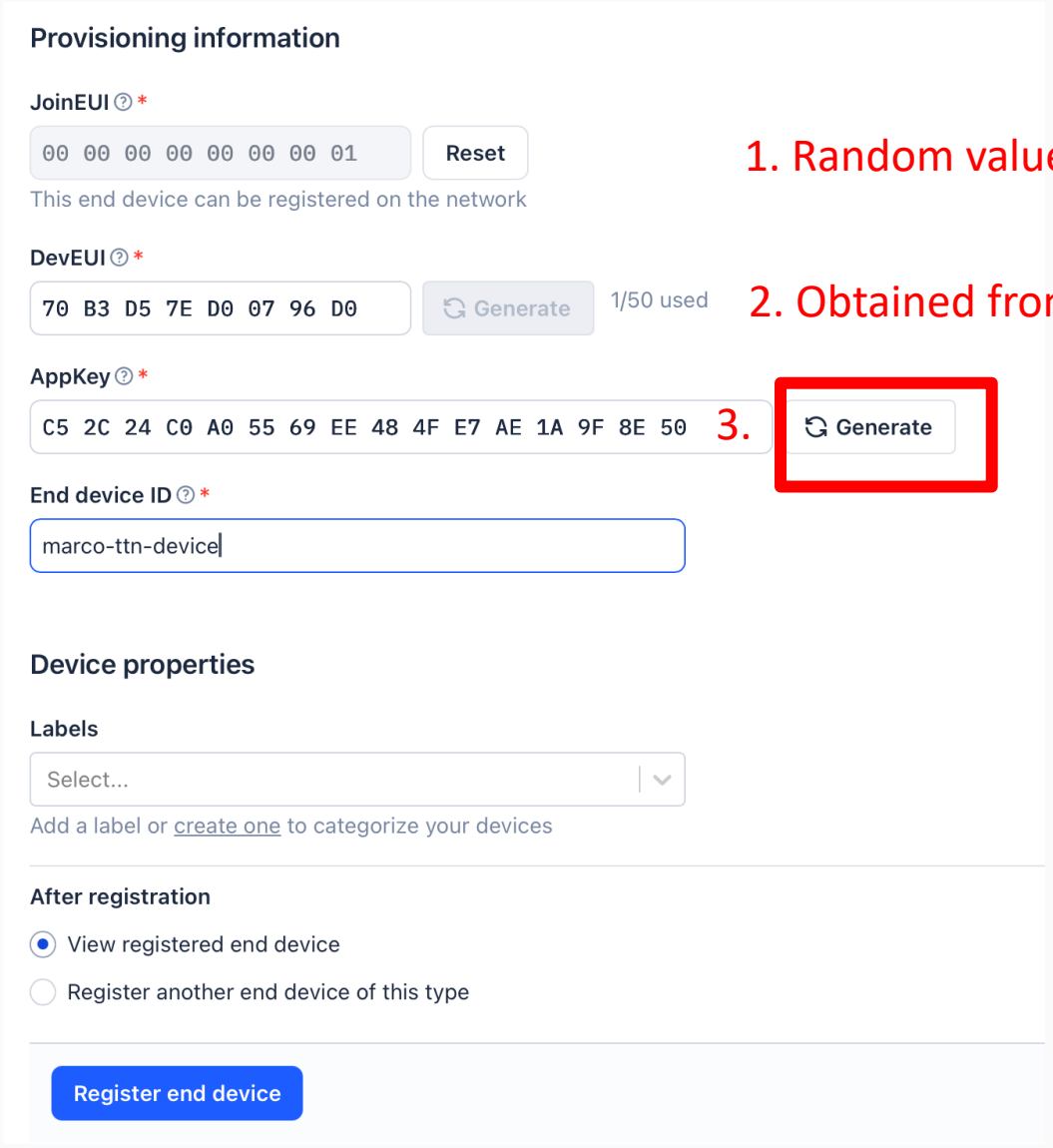
All three values go in the device sketch and must match the end device registered in TTN.

Orphaned devices

Upload the code **nicla_ttn_lora_setup.ino** to set the new JoinEUI and to visualize the DevEUI.



Lab


Provisioning information
JoinEUI *
00 00 00 00 00 00 00 01
Reset
This end device can be registered on the network
1. Random value
DevEUI *
70 B3 D5 7E D0 07 96 D0
Generate
1/50 used
2. Obtained from device
AppKey *
C5 2C 24 C0 A0 55 69 EE 48 4F E7 AE 1A 9F 8E 50
3.
Generate
End device ID *
marco-ttn-device
Device properties
Labels
Select...
Add a label or create one to categorize your devices
After registration
View registered end device
Register another end device of this type
Register end device

Lab

The Arduino sketch that sends data via LoRaWAN is called **nicla-ttn-lora.ino**

The Nicla Vision uses the Seeed Grove LoRa-E5 as an external LoRaWAN modem. The communication between the Nicla Vision and the LoRa-E5 is done through the Serial1 UART interface by sending AT commands.

The payload sent by this sketch is a counter encoded as an 8-character hexadecimal ASCII string.

Lab

The sketch performs the following operations:

1. initializes the USB serial port, used to print debug messages on the Arduino Serial Monitor;
2. initializes Serial1 at 19200 baud, used to communicate with the LoRa-E5 module;
3. sends basic AT commands to configure the LoRa-E5 module;
4. sets the LoRaWAN operating mode to OTAA;
5. configures the AppKey generated in TTN;
6. sends the AT+JOIN command to join the LoRaWAN network;
7. waits for the response +JOIN: Network joined ;
8. periodically sends an uplink message using the AT+MSG command.

Lab

Each application has its own uplink **payload decoder**.

Our example Arduino sketch sends a counter as an 8-character hexadecimal ASCII string. Example payload sent by the device:

0000000F

Because the payload is transmitted as ASCII characters, TTN receives 8 bytes. Each byte corresponds to one ASCII character of the hexadecimal string.

TTN receives 8 bytes, corresponding to the ASCII characters:

'0' '0' '0' '0' '0' '0' '0' 'F'

The JavaScript decoder has the task of converting these 8 ASCII bytes back into a readable value.

Lab

The decoder performs three operations:

1. checks that the payload contains exactly 8 bytes;
2. converts the 8 ASCII bytes into a string;
3. converts the hexadecimal string into an integer value.

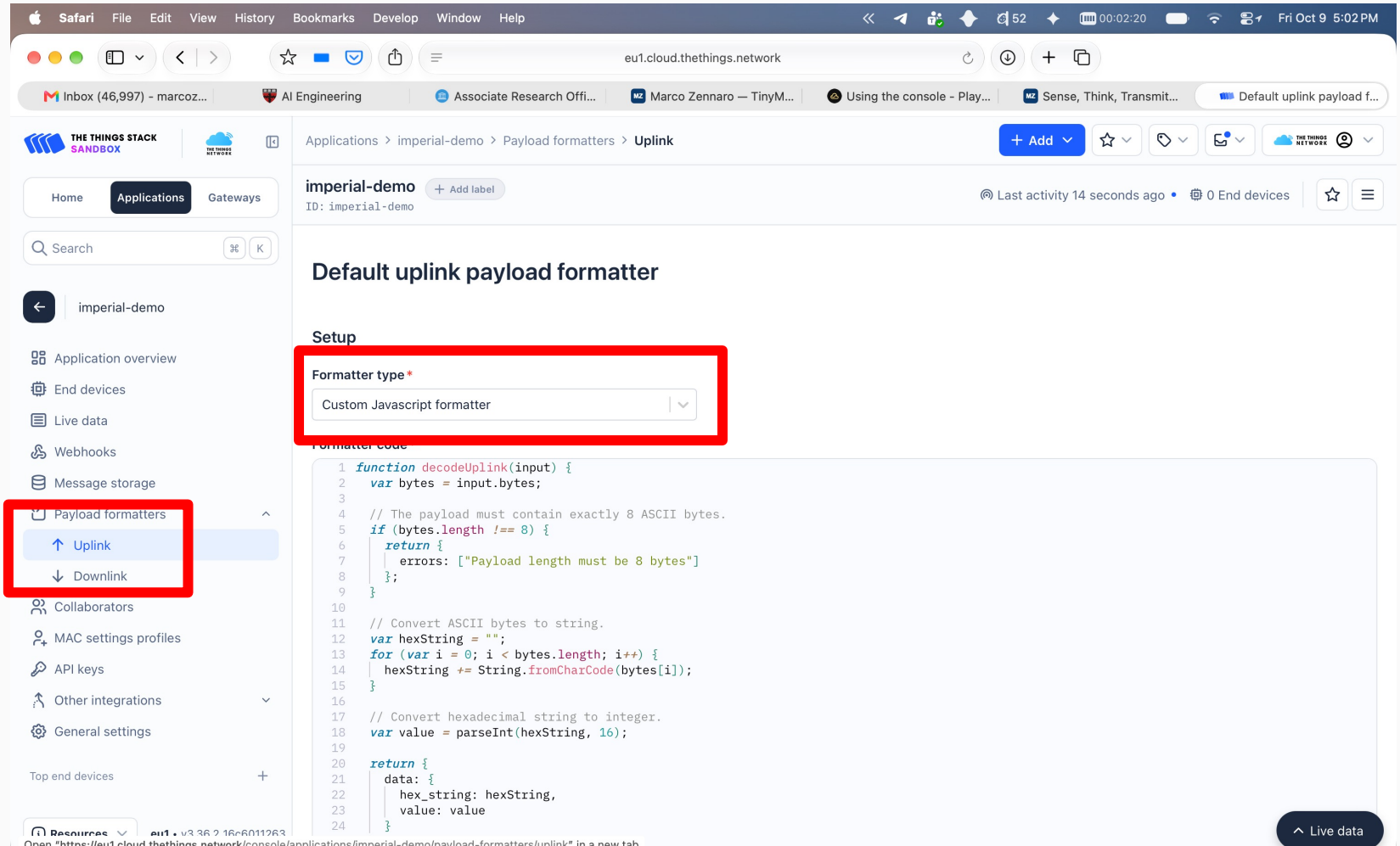
Example:

```
Input payload:  0000000F
Hex string:     "0000000F"
Decoded value:  15
```

This is why the Arduino sketch and the TTN decoder must use the same payload format!

Lab

The payload decoder for our specific application is in the file called **decoder.txt**



Lab

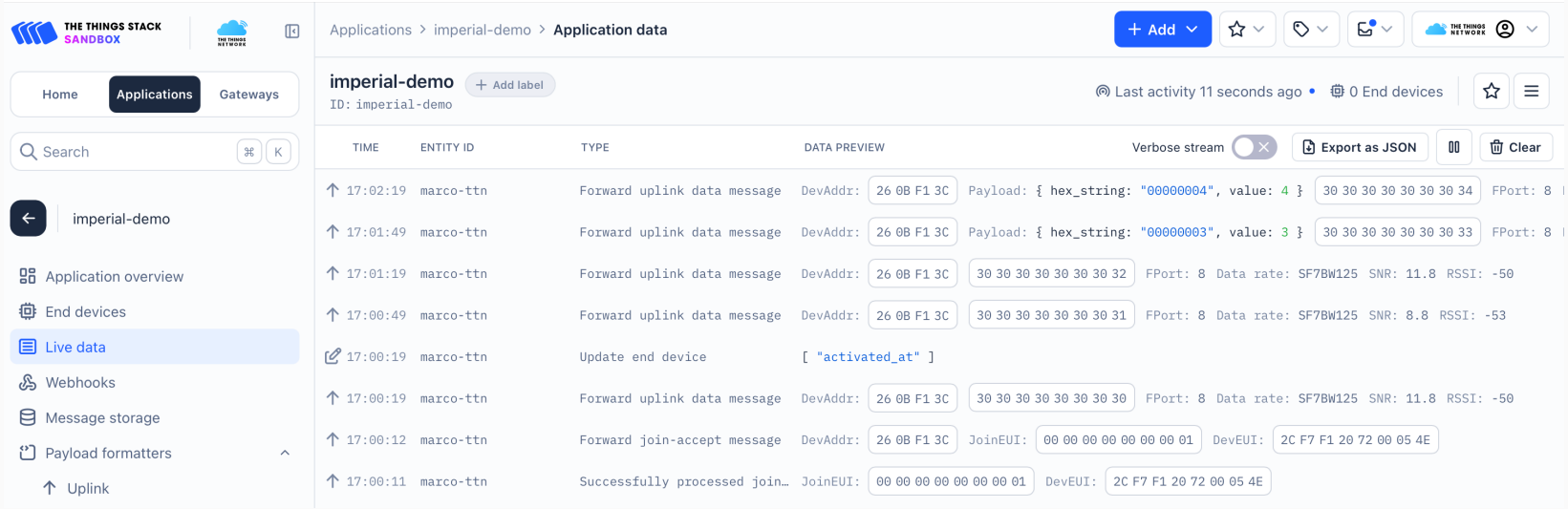
To run the Arduino code, you need to modify the AppKey configuration section in the sketch with the value generated by TTN:

```
// Configure APPKEY
// device lorae5-01 configuration:
// JoinEUI (the AppEUI obtained before): 8000000000000006
// DevEUI (obtained before): 70B3D57ED0077307
// AppKey (generated in TTN application):
00112233445566778899AABBCCDDEEFF
sendCommand("AT+KEY=APPKEY,\"00112233445566778899AABBCCDDEEFF\"");
```

Lab

You are now ready to upload the sketch!

After the upload, open the TTN application and check the live data or the device activity. The device should join the network and send the first uplink message.

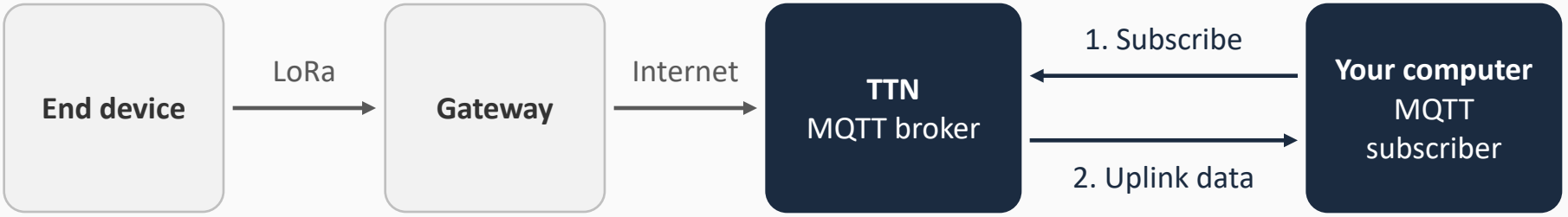


Lab

To visualize the data sent to TTN you can use MQTT.

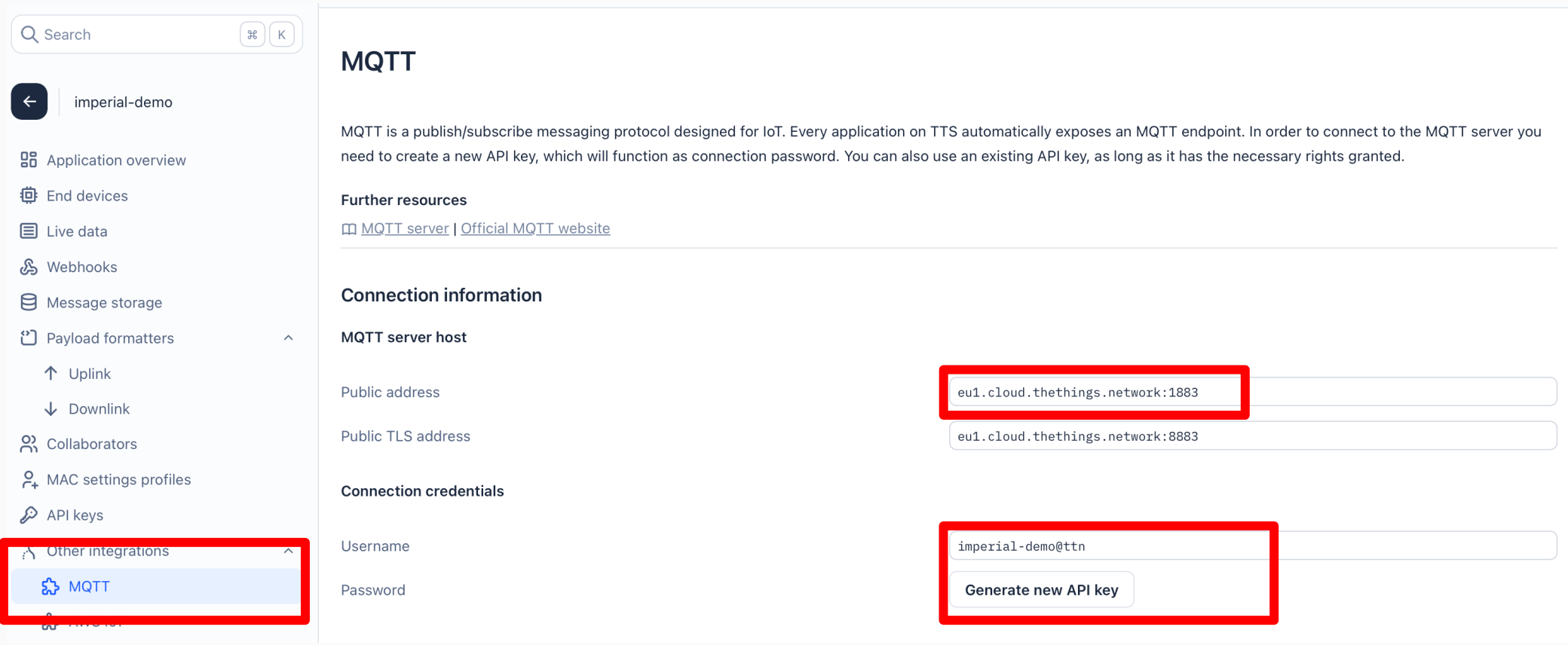
The TTN application acts as an MQTT broker and you can subscribe to the data that is being sent.

For privacy and security reasons, you must connect to the broker (TTN) with a username and password. This information is available in the TTN integrations menu.



Lab

For our example application, here are the data to connect:



Warm-up Task



Lab

Exercise:

Setup your Nicla and visualize the data sent by the Nicla using MQTT on your smartphone

Primary Task



Lab

Exercise:

Try to send something different than the counter and write a payload decoder to decode it. Don't forget that the format in the Arduino sketch and in the decoder must correspond

How far can you go with the device? Try moving outdoors!

References

- https://www.semtech.com/lora
- https://www.thethingsnetwork.org/docs/lorawan/
- https://docs.rakwireless.com/Knowledge-Hub/Learn/
- https://www.univ-smb.fr/lorawan/en/
- State of LoRaWAN and the IoT Ecosystem by Wienke Giezeman
  - https://www.youtube.com/watch?v=f3fxpsSxQzY











Thanks!

Thanks!

TinyML whispers low,
Code dances on micro souls,
AI blooms in small.